



## Full Length Article

## Context-dependent roles of familiarity, discipline, and personality in shaping social engagement in cooperative immersive learning

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## ABSTRACT

Collaborative learning represents a fundamental component of higher education, yet disparities in disciplinary perspectives, communication styles, and pre-existing social ties can lead to divergent interpretations of tasks and unbalanced contribution dynamics. Immersive virtual reality (IVR) offers innovative opportunities to mitigate these challenges. This study examines the combined influence of interpersonal familiarity, academic discipline, and personality trait composition on social interaction within a cooperative two-player IVR escape room game. A mixed methods approach, which combined quantitative questionnaires with qualitative observations and interviews, involved 108 students who formed 54 pairs. The analysis reveals a complex, context-dependent relationship between familiarity and participation. The study shows that although familiarity is generally associated with greater confidence and social interdependence, unfamiliar interdisciplinary pairs demonstrate significantly greater behavioral involvement, suggesting that unfamiliarity and cognitive divergence may be linked to more active communication and exploration. Furthermore, personality similarity was significantly associated with the satisfaction of unfamiliar pairs, and unfamiliar extrovert–extrovert pairs reported the highest motivational satisfaction. These findings challenge the assumption that familiarity is universally beneficial and offer actionable strategies for composing student groups to align with specific pedagogical objectives, such as pairing unfamiliar interdisciplinary students to promote exploration or familiar homogeneous groups to maximize efficiency. These findings provide an improved framework for collaboration, highlighting the need to align team composition with task structure in the design of cooperative IVR learning environments.

## 1. Introduction

Collaborative learning represents a fundamental component of higher education, promoting cognitive engagement, critical thinking, and interpersonal competencies through the co-construction of knowledge (Ramdani et al., 2022). However, contemporary educational settings are defined by interdisciplinary curricula, sizable cohorts, and culturally diverse classes, factors that significantly complicate the dynamics of collaboration. Disparities in disciplinary perspectives, communication styles, and pre-existing social ties can lead to divergent interpretations of tasks and unbalanced contribution dynamics (Wyman & Watson, 2020). Such inequalities, when not strategically managed, can reduce engagement, decrease motivation, and make learning ineffective, highlighting the need for data-driven methods to improve collaborative structures (Nguyen & Duong, 2025).

Immersive virtual reality (IVR) presents innovative opportunities to mitigate these challenges by allowing spatially organized tasks that increase social presence and immersion (Srivastava & Chandra, 2018). Multi-user IVR settings allow for fine-grained experimental manipulation of interaction variables and task design (Checa et al., 2023), facilitating a detailed analysis of collaborative mechanisms. Game-based IVR scenarios, such as escape rooms, further enrich problem-solving contexts by embedding information asymmetry and time-sensitive cooperation (Veldkamp et al., 2020), creating ideal conditions for examining how social variables shape engagement.

Existing studies have examined three key factors in isolation: interpersonal familiarity (increasing trust but potentially limiting novelty) (Tamimy et al., 2023); disciplinary diversity (promoting creativity while introducing coordination challenges) (Wyman & Watson, 2020);

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and personality composition (where extraversion often correlates with greater engagement) (Putriana, 2020). Importantly, it remains unclear how interpersonal familiarity interacts with the disciplinary composition and, separately, with the personality composition to shape social interaction, particularly in highly interdependent collaborative tasks. This constitutes a significant gap, as prior studies have typically assessed these factors separately in conventional educational settings (Li et al., 2023), overlooking the unique mediating role of IVR. Taken together, previous studies have addressed the independent contributions of familiarity, disciplinary diversity, and personality composition to collaboration, but have not explored how these factors interact within a single high-interdependence IVR setting. Furthermore, while theoretical frameworks such as socially shared regulation of learning (SSRL) (Sharma et al., 2024) and self-determination theory (SDT) (Deci & Ryan, 2012) explain collaborative processes, they rarely treat team composition as a central, multi-faceted independent variable within immersive environments.

The novelty and theoretical importance of this study lie in its integrative and multifactorial design, which examines how familiarity, disciplinary diversity, and personality compatibility jointly shape collaboration within a single controlled cooperative IVR experiment. By exploring these factors and their interactions under the high-interdependence conditions of IVR, where social and cognitive processes are intensified, this study directly tests the core group-composition and collaboration theories and goes beyond one-factor accounts. This integrated design enables direct comparisons of multiple social-composition factors within a shared experimental context, rather than across separate or only loosely comparable settings. The goal is to refine collaboration models by determining when and how familiarity interacts with academic discipline and personality traits, challenging the assumption that familiarity is uniformly beneficial, and identifying context-dependent trade-offs. The resulting insights will inform evidence-based strategies for forming student teams to meet specific educational goals (e.g., pairing unfamiliar, interdisciplinary students for exploration versus familiar, similar groups for efficiency). To this end, this study addresses the following research questions (RQs):

- RQ1** How does pairing unfamiliar students from different faculties impact social interaction within a two-player cooperative IVR game?
- RQ2** How does pairing unfamiliar students with mismatched personality traits impact social interaction within a two-player cooperative IVR game?

The objective of this study is to examine how interpersonal familiarity, academic discipline, and personality composition jointly shape collaboration in cooperative IVR. Specifically, this study aims to: (a) examine the interaction between familiarity and discipline; (b) examine the interaction between familiarity and personality traits; and (c) derive pedagogically actionable group-composition strategies for cooperative IVR learning.

The remainder of the paper is organized as follows. Section 2 reviews related work; Sections 3 and 4 describe the methodology and experimental design; Section 5 presents the results; Section 6 discusses the findings and limitations; and Section 7 concludes the paper.

## 2. Related work

The impact of team familiarity on IVR collaboration shows context-dependent effects. Fernandez-Espinosa et al. (2025) demonstrated that unfamiliar teams in social VR could establish trust and closeness comparable to familiar ones, suggesting that the medium can bypass offline social preferences. In contrast, Tütüncü et al. (2025) found that established pairs communicated more fluidly, although their focus on pairs leaves open questions about larger, interdependent teams. Supporting the notion that unfamiliarity is not inherently disadvantageous, Karaosmanoglu et al. (2023) found that unfamiliar teams performed and

communicated as effectively as familiar ones in an asymmetric task, with shifting roles sometimes increasing participation. These mixed findings highlight that the advantages of familiarity are not universal and require further systematic investigation.

Disciplinary heterogeneity is another important dimension. Studies in healthcare simulations, such as those of Neher et al. (2024) and Williams et al. (2020), show that interprofessional cohorts can improve understanding of mutual roles and communication efficiency. However, Alghanaim et al. (2025) reported the challenge of uneven technical proficiency, and Nikolic et al. (2019) reported the challenge of friction in the initial workflow. Even though Wyman and Watson (2020) showed that heterogeneous groups can perceive greater relevance to tasks, homogeneous groups benefit from shared epistemic frameworks. Direct comparisons are difficult due to inconsistent metrics between studies, highlighting the need for a controlled examination in a unified VR context.

Individual personality traits, particularly extraversion, influence social presence and participation in the IVR, although the findings are based on the task. Thorp et al. (2023) linked extraversion to social presence and skill transfer, but not to direct performance metrics. Markowitz et al. (2025) found that extraversion affected group satisfaction and cohesion, respectively. The composition of personality within a group also matters: Zhu et al. (2024) found that groups with lower average agreeability benefited the most from collaboration, and Mueller et al. (2025) showed that extraversion heterogeneity could stabilize performance. However, most of the evidence comes from low-interdependence settings, which limits insights for tightly coupled IVR collaboration.

Several interventions aim to increase interaction within unfamiliar or interdisciplinary VR teams. Neher et al. (2024) and Williams et al. (2020) indicated that structured role assignment and formalized handover protocols improved communication efficiency and self-efficiency among mixed-discipline groups. Hung et al. (2024) and Li et al. (2023) demonstrated that socially shared regulation tools and collaboration scripts were shown to improve monitoring, discourse management, and joint planning, reinforcing participation. The design of social signals also proves impactful; Li et al. (2025) showed that avatars in collaborative virtual environments can synchronize joint attention and strengthen peer presence, while Kim et al. (2016) suggested that proactive virtual agents increased perceived social presence between various user personalities. Building on this, Knutzen et al. (2025) showed that cross-cultural modules in social VR improved group efficiency and experiences, but most interventions alter tasks or add supports instead of isolating the effects of disciplinary composition or its interaction with familiarity.

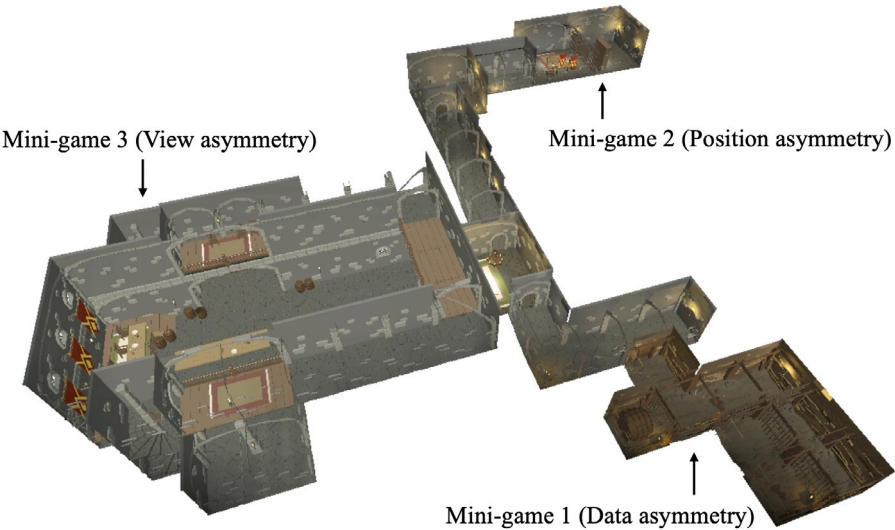
In summary, while familiarity, disciplinary background, and personality independently shape collaborative dynamics, their combined effects on interdependent IVR tasks remain unclear. Existing research often examines these factors in isolation or in low-stakes environments. This study addresses this gap by systematically examining how familiarity interacts with disciplinary composition (intra- vs. inter-faculty pairs) and, separately, with personality pairing (matched vs. mismatched traits) during a collaborative IVR puzzle-solving task. Table 1 provides a summary of existing studies and outlines how the present study addresses the identified gaps.

## 3. Methods

This study uses mixed methods, quantitative questionnaires, and qualitative observations to examine how familiarity, discipline, and personality affect social interaction in a two-player IVR environment. A custom cooperative VR escape room served as both the intervention and the data source. The following subsections detail the rationale for collaborative design and the guiding theory of the IVR game.

**Table 1**  
Summary of existing studies and how the present study addresses identified gaps.

| Content                  | Existing studies                                                                                                                                                                                                                             | Gap and the present study                                                                                                                                                                                   |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Familiarity              | Prior VR research indicates that familiarity influences trust, communication, and participation in collaborative contexts (Fernandez-Espinosa et al., 2025; Karaosmanoglu et al., 2023; Tütüncü et al., 2025).                               | Familiarity has largely been explored in isolation. The present study investigates it jointly with disciplinary background and personality composition within a single cooperative IVR setting.             |
| Disciplinary composition | Interdisciplinary groups may enhance role understanding and task relevance, but they can also increase coordination demands (Alghanaim et al., 2025; Neher et al., 2024; Nikolic et al., 2019; Williams et al., 2020; Wyman & Watson, 2020). | Evidence rarely derives from IVR contexts and is difficult to compare across settings. The present study examines the interaction between familiarity and disciplinary background within a shared IVR task. |
| Personality composition  | Extraversion and group-level personality composition have been linked to social presence, satisfaction, and collaboration outcomes (Markowitz et al., 2025; Mueller et al., 2025; Thorp et al., 2023; Zhu et al., 2024).                     | Their role in tightly interdependent IVR collaboration remains unclear. The present study examines the interaction between familiarity and personality composition in a cooperative IVR task.               |
| IVR support              | Prior IVR research demonstrates that scripts, roles, avatars, and agents can improve collaboration (Hung et al., 2024; Kim et al., 2016; Knutzen et al., 2025; Li et al., 2023, 2025).                                                       | These studies primarily target task support rather than team composition. The present study instead foregrounds team composition in a controlled IVR experiment.                                            |



**Fig. 1.** Overview structure of the two-player cooperative IVR game, showing three mini-games each featuring a distinct collaborative asymmetry: data (mini-game 1), position (mini-game 2), and view (mini-game 3).

3.1. Collaborative asymmetrical integration

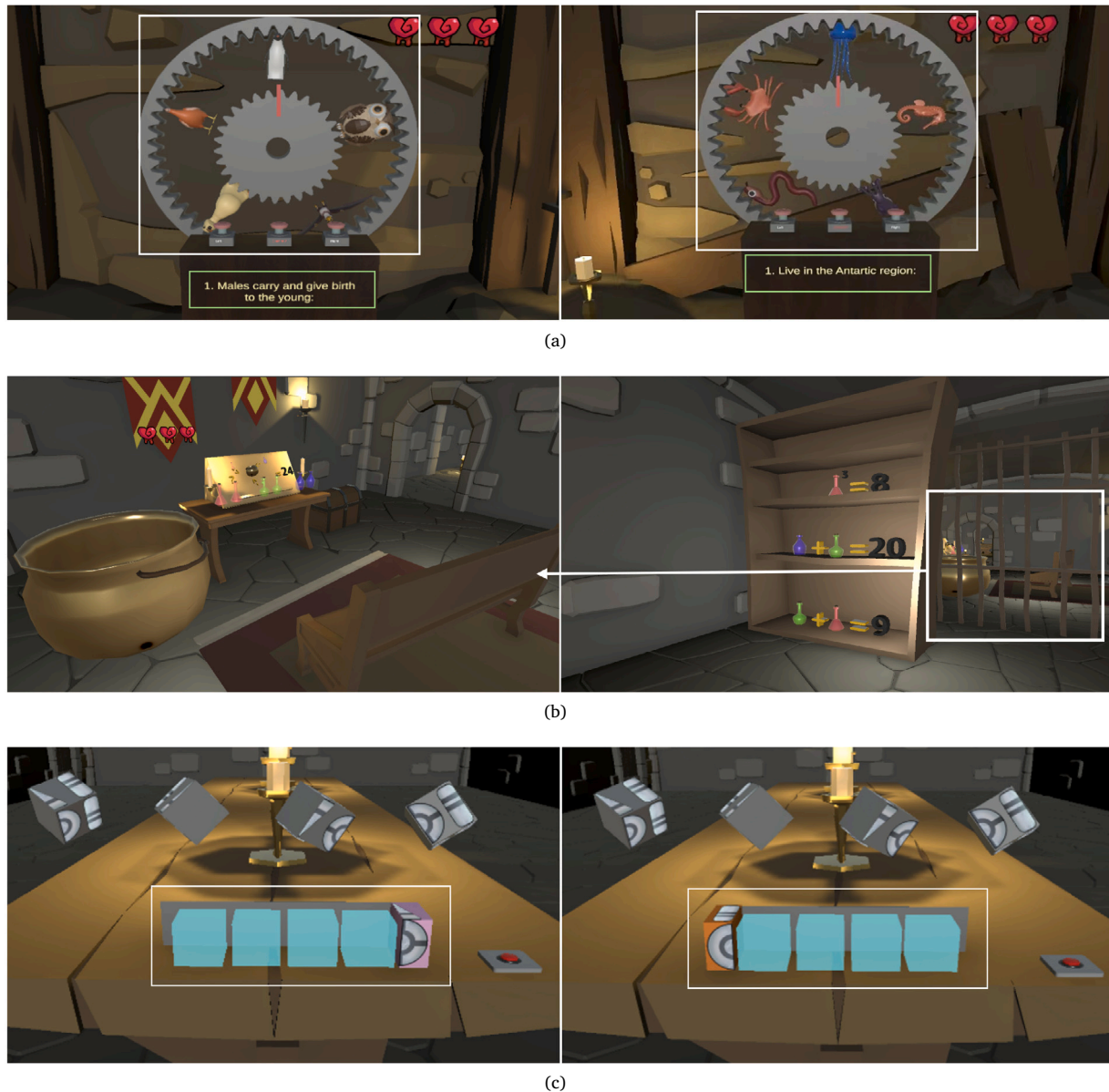
To investigate how pairs composition shapes collaboration under high interdependence, this study designed a two-player IVR escape room. Fig. 1 shows the core design principle, the systematic implementation of three collaborative asymmetries: data, position, and view, derived from theoretical frameworks for team coordination. These asymmetries ensure non-shared information and impose high coordination demands, thus creating the strong interdependence necessary to examine our RQs concerning familiarity, disciplinary background, and personality. Each of the three sequential mini-games isolates one primary asymmetry, which requires active communication and interdependent reliance to solve puzzles. The three mini-games were completed in a fixed sequence and typically finished in approximately 30 min; no explicit per-mini-game or general in-game time limit was imposed. No in-game hint mechanism was provided, and when participants failed a puzzle, they restarted the corresponding mini-game from the beginning. Task difficulty was calibrated through small-scale pilot testing with different groups of students prior to the formal study.

The virtual environment was developed in Unity3D Editor v2022.3.12f1c1 LTS, managed via Unity Hub v3.3.0-c9 (Unity Technologies, 2024a), using the XR Interaction Toolkit v2.5.2 (Unity Technologies, 2024b) together with XR Plugin Management v4.4.0, OpenXR

Plugin v1.8.2, XR Core Utilities v2.1.1, and XR Legacy Input Helpers v2.1.10 for locomotion and object interaction, and Photon Unity Networking 2 (PUN2) v2.45 (Photon Engine, 2024) for synchronized multi-user functionality. The experience was implemented on two head-mounted Meta Quest 3 displays in a standalone mode (non-PC-tethered) (Meta Platforms, Inc., 2024). The devices were configured for inside-out, room-scale six-degree-of-freedom (6-DOF) tracking. Participants interacted with the virtual environment using standard Meta Quest Touch Plus controllers.

3.1.1. Data asymmetry mini-game (Mini-game 1)

This mini-game operationalizes *data asymmetry*, where each player possesses unique, complementary information essential for the teammate’s sub-task. This design creates an information-gradient that forces verbal exchange. In the puzzle “safety locker”, Participant A (PA) sees clue fragments for the combination lock dials that Participant B (PB) must physically manipulate and vise versa. For example, PA’s clue interface displayed “Males carry and give birth to young”, which PA then communicates to PB. PB then rotates the dial to set the seahorse to point north (see Fig. 2(a)). They must synthesize these partial clues to determine the correct dial settings over three consecutive rounds. By necessitating active sharing of non-redundant information,



**Fig. 2.** Collaborative asymmetry puzzles for participant A (left) and participant B (right): (a) Mini-game 1 (data asymmetry) displays; (b) Mini-game 2 (position asymmetry) displays; and (c) Mini-game 3 (view asymmetry) displays. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

this task serves as an icebreaker and establishes a baseline of essential communication.

### 3.1.2. Position asymmetry mini-game (Mini-game 2)

This mini-game implements *positional asymmetry*, where physical access to critical resources is mutually exclusive, requiring coordinated action. The PA is stationed outside a chamber, controlling a floor trigger that temporarily opens its door. PB can enter only while PA holds the trigger, but the door closes if PA releases it, preventing PA from entering. Inside, PB finds a mathematical logic puzzle (see Fig. 2(b)). The numerical solution must be communicated to PA, who uses it to identify and place the correspondingly labeled potion bottles in a cauldron in the outer area. This spatial and functional division prompts strategic planning about when to activate the door and how to collaboratively solve the embedded puzzle under access constraints.

### 3.1.3. View asymmetry mini-game (Mini-game 3)

This mini-game creates *view asymmetry*, where each player has a unique, fixed perspective on a shared object, necessitating the integration of partial perceptions to form a complete mental model. The goal is to insert and rotate four cubic blocks into a frame to assemble a complete four-quadrant image. PA sees only the left half of each block's faces, while PB sees only the right half. To correctly position the blocks in the blue area, they must describe the visible patterns (see Fig. 2(c)) and collaboratively decide how to rotate the blocks so that their orientations and positions match those shown on the left and right halves of the block faces. This task emphasizes perspective-taking, precise verbal description, and collaborative mental synthesis.

### 3.2. Design rationale and theoretical alignment

The design of the three collaborative IVR mini-games is theoretically grounded in the challenges of asymmetric information and coordinated

**Table 2**

Social interaction-driven game design method and motivators from the Octalysis framework (Mohanty & Christopher, 2023).

| Core drive       | Method                   | Description                                                                                                                |
|------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Social influence | Team quests              | Enforces cooperative through interactive puzzle-solving, encouraging meaningful team engagement.                           |
| Unpredictability | Visual storytelling      | Sparks curiosity and enables team discussions via visual cues in puzzles.                                                  |
| Accomplishment   | Quest lists, Aura effect | Encourages effective teamwork, idea sharing, communication, and critical thinking through collaborative puzzle completion. |
| Epic meaning     | Narrative                | Uses friendship-themed narratives to inspire teamwork and facilitate dungeon escape.                                       |
| Empowerment      | Dynamic feedback         | Motivates persistent problem-solving through verbalized thoughts, feedback exchange, and improved communication.           |

action. Each game systematically manipulates a specific dimension of asymmetry, including information (data), action (position), and perspective (view), to create a controlled escalation of coordination demands. This progression directly serves the core research objective of probing how pairs characteristics influence the social and cognitive processes required to overcome non-shared information. The data game establishes a baseline for information exchange, the position game introduces demands for coordinated physical action under constraint, and the view game integrates perceptual asymmetry, requiring teams to synthesize fundamentally different viewpoints. This structure provides a robust platform to examine joint monitoring, communication clarity, and co-regulation.

This study uses a game design method drawn from the Octalysis framework (Mohanty & Christopher, 2023) to improve collaborative interactions between participants with different academic disciplines and personality traits. The design prioritizes techniques that promote clear communication, cooperative problem-solving, and sustained mutual engagement. Table 2 provides an overview of the design method used to improve social interactions.

The design utilizes the core drives of the Octalysis framework, particularly the *White Hat* motivators *accomplishment*, *meaning*, and *empowerment* (Weber et al., 2022), to maintain participant involvement and promote positive psychological results during collaborative problem-solving. Structured puzzles that require joint effort activate the *accomplishment* drive, cultivating a sense of achievement that increases self-confidence and willingness to collaborate. The *meaning* drive extends the experience of participants beyond mechanical task completion by embedding a narrative that unites them around shared objectives and collective successes. The *empowerment* enables participants to influence the progression of the game through cooperative decision-making, thus strengthening both the *accomplishment* and the productive team collaboration. Furthermore, the design integrates intrinsic motivators in the right brain, *empowerment*, *social influence*, and *unpredictability* (Kian et al., 2022) to promote social interaction through inherent cooperative enjoyment and alignment with self-directed learning theories. Through combining *White Hat* and intrinsic motivators, this approach aims to support individual progression and team success, improve social interaction, and cultivate collective achievement.

## 4. Experiment design

### 4.1. Participants

The study recruited 108 undergraduate and postgraduate students aged 19 to 26 years ( $\mu_{age} = 21.83$ ,  $\sigma_{age} = 1.66$ ) recruited through open advertisements. The participants were assigned to one of two familiarities:

**Table 3**

Overview of experimental design, including number of pairs by familiarity levels, academic discipline pairing, and personality trait matching.

| Familiarity  | Discipline pairing | Personality matching | Number of pairs |
|--------------|--------------------|----------------------|-----------------|
| Familiar     | Intra-Faculty      | Matched              | 8               |
| Familiar     | Intra-Faculty      | Mismatched           | 4               |
| Familiar     | Inter-Faculty      | Matched              | 6               |
| Familiar     | Inter-Faculty      | Mismatched           | 9               |
| Unfamiliar   | Intra-Faculty      | Matched              | 10              |
| Unfamiliar   | Intra-Faculty      | Mismatched           | 3               |
| Unfamiliar   | Inter-Faculty      | Matched              | 9               |
| Unfamiliar   | Inter-Faculty      | Mismatched           | 5               |
| <b>Total</b> |                    |                      | <b>54</b>       |

**Familiar pairs** are self-identified friends with a relationship of at least 6 months who engage in shared activities three or more times a week.

**Unfamiliar pairs** are randomly matched individual registrants with no previous acquaintance.

Participants who did not meet these criteria were excluded. To investigate RQ1, these pairs were further categorized by academic discipline and personality trait. Disciplinary diversity was determined by enrollment in one of the three faculties (Business, Science & Engineering or Social Sciences). Personality (introversion or extroversion) was assessed using the Eysenck Personality Questionnaire (EPQR-A) (Francis et al., 1992). Table 3 summarizes demographics by familiarity, academic discipline, and personality trait, with Tables A.1 and A.2 providing detailed distributions for familiar and unfamiliar pairs, respectively.

When forming pairs within the familiarity conditions, this study deliberately aimed for a balanced distribution of intra- and inter-faculty pairs, as well as personality-matched (introvert–introvert; extrovert–extrovert) and mixed pairs. This strategic sampling was implemented to ensure that no single subgroup dominated the sample, allowing for more comprehensive comparisons between the factors of interest.

### 4.2. Experiment setup

Fig. 3 illustrates the experimental setup, which consisted of pre-game, gameplay, and post-game phases, together with a structured observation protocol. Each experiment lasted approximately one hour. In the pre-game phase, participants first attended an introductory briefing that outlined the objectives of the experiment. Subsequently, the participants signed a consent form acknowledging their voluntary and uncompensated participation. Next, each participant completed a 15-minute VR tutorial covering controller operation and virtual object manipulation to establish a consistent level of operational proficiency and necessary gameplay skills.

In the gameplay phase, two trained observers used a non-participatory protocol (Sassoli de Bianchi, 2013) to code the pairs' social interactions across nine dimensions (see Section 4.3.2). To ensure coding reliability, both observers underwent a calibration procedure prior to data collection, which involved a joint review of the coding scheme adapted from López-Faican and Jaén (2020), followed by discussions with all researchers to establish concrete behavioral examples and consistent decision rules for each dimension. During the 30-minute session, observers documented participants' physical gestures and verbal communication patterns from a non-intrusive distance within the same room. Participants were advised of their right to withdraw at any time. In the post-game phase, a brief rest period preceded the completion of individual questionnaires and one-on-one feedback interviews.

This study was reviewed and approved by University of Nottingham Ningbo China (UNNC) Research Ethics Committee, dated 30 October 2023. Participation was voluntary and written informed consent was obtained from all individual participants included in this study.

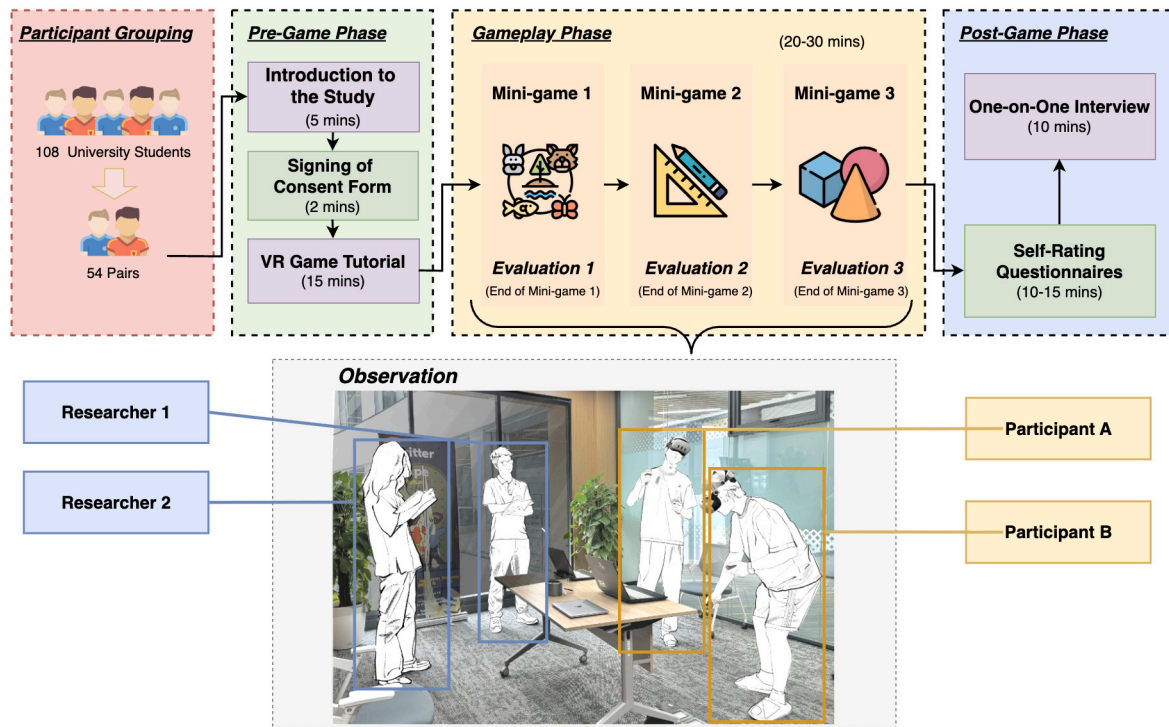


Fig. 3. Overview of the experimental procedure, detailing the pre-game briefing and tutorial, the gameplay phase with non-participatory observation, and post-game interviews along with self-rating questionnaires.

#### 4.3. Data collection and measurements

This study used a mixed-methods approach, combining quantitative questionnaires with qualitative observations and interviews. This design provided a comprehensive understanding of how the cooperative IVR game influenced social engagement, particularly with respect to differences in academic discipline (RQ1) and personality trait (RQ2).

**Quantitative data** from self-report questionnaires offered standardized metrics on social interaction, motivation, and collaboration during gameplay.

**Qualitative data** from behavioral observations and interviews captured real-time social behaviors and explored participants' subjective experiences, communication dynamics, and sense of social connection.

Based on the RQs, the following null hypotheses were tested:

- **H0<sub>1</sub>**: There are no statistically significant differences in social interaction between pairs of unfamiliar students from the same faculty and pairs from different faculties during a cooperative two-player IVR game.
- **H0<sub>2</sub>**: There are no statistically significant differences in social interaction between pairs of unfamiliar students with matched personality traits and pairs with mismatched personality traits during a cooperative two-player IVR game.

##### 4.3.1. Self-rating questionnaires

The standardized questionnaires provided quantitative metrics on the collaborative experience. The **Social Presence Module of the Game Experience Questionnaire** (GEQ-SPM) (Norman, 2013; Zulfa et al., 2020) assessed the perceived interpersonal connection, measuring the psychological and behavioral involvement. The **Instructional Materials Motivation Survey** (IMMS) (Cardoso-Júnior & Faria, 2021), based on Keller's ARCS model (Keller, 1979), evaluated intrinsic motivation (Attention, Relevance, Confidence, Satisfaction). Collaborative

dynamics was measured using the Social Interdependence in Collaboration Scales (SOCS) (Shimizu et al., 2020), which captures positive and negative interdependence and team processing, and the **Collaboration Module of the Collaborative Learning, Social Presence and Satisfaction** (CLSS\_CM) questionnaire (So & Brush, 2008), which evaluates communication quality, trust, and mutual support.

Questionnaire responses were collected individually during the post-game phase and analyzed at the individual level rather than analyzed within pairs, as the instruments were designed to capture each participant's own subjective perceptions of social presence, motivation and collaboration during the IVR experience.

##### 4.3.2. Observation

Interaction behaviors were assessed in real time at the pair level using an observation template adapted from Marroun and Young (2024). Following each game segment, two trained observers independently rated the interactions of each pair on a five-point Likert scale in nine dimensions of cooperative work (see Table A.3). These dimensions evaluated key processes such as maintaining mutual understanding, dialog management, reaching consensus, task division, and reciprocal interaction.

##### 4.3.3. Interviews

Individual semi-structured interviews provided qualitative depth, exploring participants' subjective experiences of collaboration. A 14-question guide (see Table A.4) prompted reflection on how game mechanics influenced communication, willingness to share information, and perceived team dynamics. The questions specifically probed the impact of disciplinary differences (RQ1) and personality trait interactions (RQ2) on social engagement. The interview data was transcribed and analyzed to supplement the quantitative findings and generate practical insights to design inclusive cooperative IVR experiences.

**Table 4**

Descriptive statistics for the GEQ\_SPM, IMMS, SOCS, and CLSS\_CM questionnaires by familiarity and academic discipline.  $\mu$  indicates the mean score for each group;  $\sigma$  denotes the standard deviation; SW( $p$ ) refers to the  $p$ -value from the Shapiro–Wilk test for normality, with  $p > 0.05$  indicating a normal distribution.

| Measurement | Unfamiliar pairs |          |           |               |          |           | Familiar pairs |          |           |               |          |           |
|-------------|------------------|----------|-----------|---------------|----------|-----------|----------------|----------|-----------|---------------|----------|-----------|
|             | Intra-Faculty    |          |           | Inter-Faculty |          |           | Intra-Faculty  |          |           | Inter-Faculty |          |           |
|             | $\mu$            | $\sigma$ | SW( $p$ ) | $\mu$         | $\sigma$ | SW( $p$ ) | $\mu$          | $\sigma$ | SW( $p$ ) | $\mu$         | $\sigma$ | SW( $p$ ) |
| GEQ_SPM-EM  | 3.160            | 0.264    | 0.059     | 3.357         | 0.379    | 0.159     | 3.133          | 0.354    | 0.152     | 3.132         | 0.246    | 0.106     |
| GEQ_SPM-NF  | 0.892            | 0.345    | 0.106     | 0.964         | 0.322    | 0.070     | 0.993          | 0.381    | 0.094     | 0.842         | 0.277    | 0.102     |
| GEQ_SPM-BI  | 3.154            | 0.340    | 0.643     | 3.238         | 0.409    | 0.161     | 3.061          | 0.441    | 0.096     | 2.819         | 0.322    | 0.281     |
| IMMS-A      | 3.240            | 0.274    | 0.221     | 3.286         | 0.225    | 0.264     | 3.121          | 0.339    | 0.079     | 3.047         | 0.327    | 0.596     |
| IMMS-R      | 2.524            | 0.334    | 0.590     | 2.406         | 0.482    | 0.544     | 2.688          | 0.420    | 0.108     | 2.318         | 0.276    | 0.208     |
| IMMS-C      | 2.596            | 0.394    | 0.559     | 2.397         | 0.424    | 0.094     | 3.054          | 0.371    | 0.414     | 3.161         | 0.279    | 0.575     |
| IMMS-S      | 3.100            | 0.350    | 0.086     | 3.286         | 0.342    | 0.130     | 3.067          | 0.280    | 0.052     | 3.017         | 0.377    | 0.135     |
| SOCS        | 2.938            | 0.249    | 0.913     | 3.093         | 0.303    | 0.440     | 3.209          | 0.311    | 0.424     | 3.039         | 0.272    | 0.060     |
| CLSS_CM     | 2.971            | 0.290    | 0.204     | 3.067         | 0.244    | 0.486     | 3.133          | 0.392    | 0.254     | 3.188         | 0.321    | 0.312     |

GEQ\_SPM: EM - Empathy; NF - Negative Feelings; BI - Behavioral Involvement.

IMMS: A - Attention; R - Relevance; C - Confidence; S - Satisfaction.

**Table 5**

Two-way ANOVA of familiarity (FM), academic discipline (AD), and their interaction (FM  $\times$  AD). Only measurements with statistically significant differences are shown and the detailed results are presented in Table A.11.

| Measurement | Source         | Type III sum of squares | df | Mean square | $F$    | $p$       | $\eta^2 p$ |
|-------------|----------------|-------------------------|----|-------------|--------|-----------|------------|
| GEQ_SPM-EM  | FM             | 0.426                   | 1  | 0.426       | 4.164  | 0.044*    | 0.038      |
| GEQ_SPM-BI  | FM             | 1.753                   | 1  | 1.753       | 13.519 | <0.001*** | 0.115      |
|             | FM $\times$ AD | 0.712                   | 1  | 0.712       | 5.491  | 0.021*    | 0.050      |
| IMMS-A      | FM             | 0.861                   | 1  | 0.861       | 9.913  | 0.002**   | 0.087      |
| IMMS-R      | AD             | 1.594                   | 1  | 1.594       | 10.412 | 0.002**   | 0.091      |
| IMMS-C      | FM             | 10.013                  | 1  | 10.013      | 71.734 | <0.001*** | 0.408      |
|             | FM $\times$ AD | 0.628                   | 1  | 0.628       | 4.501  | 0.036*    | 0.041      |
| IMMS-S      | FM             | 0.613                   | 1  | 0.613       | 5.423  | 0.022*    | 0.049      |
| SOCS        | FM $\times$ AD | 0.705                   | 1  | 0.705       | 8.589  | 0.004**   | 0.076      |
| CLSS_CM     | FM             | 0.536                   | 1  | 0.536       | 5.280  | 0.024*    | 0.048      |

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

## 5. Results

### 5.1. Familiarity and academic discipline

#### 5.1.1. Descriptive and inferential statistical results

Table 4 presents the descriptive statistics for all dependent measures according to familiarity and academic discipline (see Table A.5 for details). The initial result suggests key patterns: the scores of GEQ\_SPM-EM and GEQ\_SPM-NF were stable under all conditions, while GEQ\_SPM-BI varied, with the highest mean for unfamiliar inter-faculty pairs ( $\mu = 3.238$ ) and the lowest for familiar inter-faculty pairs ( $\mu = 2.819$ ). For the IMMS, confidence (IMMS-C) was consistently higher among familiar pairs. Collaborative measures (SOCS, CLSS\_CM) also showed variation, with familiar intra-faculty pairs reporting the highest positive interdependence ( $\mu = 3.209$ ). The assumptions of normality and homogeneity of variance were verified. The Shapiro–Wilk test and Levene's test (all  $p > .05$ , see Table A.6) did not indicate violations, supporting the validity of the parametric analyses.

To assess the significance of these observed differences, two-way ANOVAs were conducted. Table 5 summarizes the ANOVA results that revealed several significant effects. Familiarity had a significant main effect on GEQ\_SPM-EM, GEQ\_SPM-BI, IMMS-A, IMMS-C, IMMS-S and CLSS\_CM ( $p < .05$ ). In contrast, academic discipline produced a significant main effect only for IMMS-R ( $F(1104) = 10.412$ ,  $p = 0.002$ ,  $\eta^2 p = .091$ ), with inter-faculty pairs reporting greater relevance. Significant interaction effects between familiarity and academic discipline were found for GEQ\_SPM-BI, IMMS-C, and SOCS ( $p < .05$ ). These interactions indicate that the influence of familiarity depended on whether pairs were intra- or inter-faculty.

#### 5.1.2. Post-hoc analysis of interaction effects

Fig. 4 presents the results of Fisher's LSD post-hoc tests conducted to examine the significant interactions. For GEQ\_SPM-BI, familiarity effects were discipline-dependent: unfamiliar inter-faculty pairs scored significantly higher than familiar inter-faculty pairs ( $\Delta = -.419$ ,  $p < .001$ ), while there were no significant differences for intra-faculty pairs. Furthermore, familiar intra-faculty pairs outperformed familiar inter-faculty pairs ( $\Delta = -.242$ ,  $p < .016$ ).

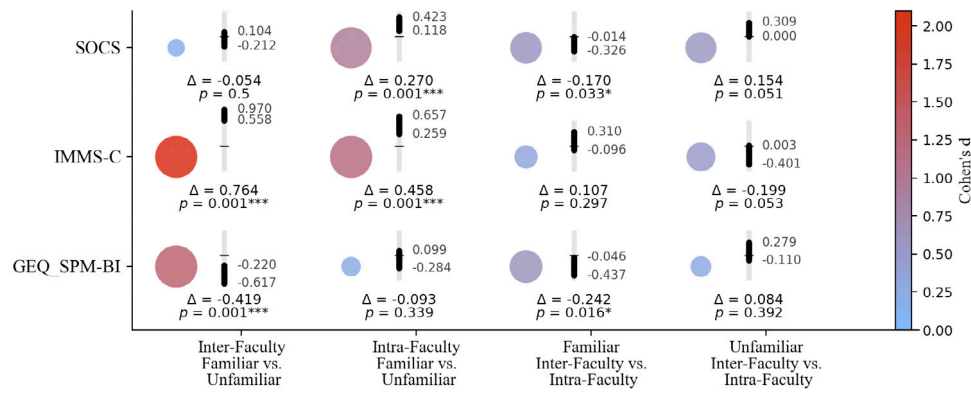
For IMMS-C, a strong main effect of familiarity was evident: familiar pairs scored higher than unfamiliar pairs under both intra-faculty ( $\Delta = .458$ ,  $p < .001$ ) and inter-faculty ( $\Delta = .764$ ,  $p < .001$ ) conditions, without a significant effect of discipline alone. The pattern for SOCS differed: familiarity significantly increased scores for intra-faculty pairs ( $\Delta = .270$ ,  $p < .001$ ) but not for inter-faculty pairs. In addition, familiar intra-faculty pairs reported higher SOCS than familiar inter-faculty pairs ( $\Delta = -.170$ ,  $p = .033$ ).

In summary, the results demonstrate that while familiarity broadly increased confidence and certain affective measures, its impact on behavioral involvement and social interdependence was dependent on disciplinary alignment. Table A.12 reports the complete detailed results.

### 5.2. Familiarity and personality trait

#### 5.2.1. Descriptive and inferential statistical results

Table 6 presents the descriptive statistics for all measures based on familiarity and combination of personality traits (introvert–introvert, extrovert–extrovert, mismatched) (see Table A.5 for details). Preliminary observations indicate patterns: GEQ\_SPM-BI was highest for unfamiliar extrovert–extrovert pairs ( $\mu = 3.285$ ) and lowest for familiar



**Fig. 4.** Fisher's LSD post-hoc pairwise comparisons for the interactions between familiarity and academic discipline on the measures of GEQ\_SPM-BI, IMMS-A and IMMS-S.  $\Delta$  represents the mean difference between groups;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Table 6**

Descriptive statistics for the GEQ\_SPM, IMMS, SOCS, and CLSS\_CM questionnaires by familiarity and personality trait.  $\mu$  indicates the mean score for each group;  $\sigma$  denotes the standard deviation; SW( $p$ ) refers to the  $p$ -value from the Shapiro-Wilk test for normality, with  $p > 0.05$  indicating a normal distribution.

| Measurement | Unfamiliar pairs |          |           |       |          |           |            |          |           | Familiar pairs |          |           |       |          |           |            |          |           |
|-------------|------------------|----------|-----------|-------|----------|-----------|------------|----------|-----------|----------------|----------|-----------|-------|----------|-----------|------------|----------|-----------|
|             | I-I              |          |           | E-E   |          |           | Mismatched |          |           | I-I            |          |           | E-E   |          |           | Mismatched |          |           |
|             | $\mu$            | $\sigma$ | SW( $p$ ) | $\mu$ | $\sigma$ | SW( $p$ ) | $\mu$      | $\sigma$ | SW( $p$ ) | $\mu$          | $\sigma$ | SW( $p$ ) | $\mu$ | $\sigma$ | SW( $p$ ) | $\mu$      | $\sigma$ | SW( $p$ ) |
| GEQ_SPM-EM  | 3.194            | 0.401    | 0.403     | 3.222 | 0.302    | 0.078     | 3.361      | 0.344    | 0.852     | 3.028          | 0.211    | 0.281     | 3.167 | 0.256    | 0.713     | 3.160      | 0.376    | 0.062     |
| GEQ_SPM-NF  | 0.800            | 0.270    | 0.354     | 1.008 | 0.336    | 0.293     | 0.911      | 0.351    | 0.199     | 0.817          | 0.301    | 0.161     | 0.967 | 0.365    | 0.223     | 0.950      | 0.350    | 0.297     |
| GEQ_SPM-BI  | 3.111            | 0.351    | 0.919     | 3.285 | 0.305    | 0.100     | 3.139      | 0.320    | 0.199     | 2.792          | 0.349    | 0.626     | 2.843 | 0.385    | 0.124     | 3.118      | 0.404    | 0.067     |
| IMMS-A      | 3.271            | 0.219    | 0.075     | 3.323 | 0.255    | 0.099     | 3.181      | 0.247    | 0.803     | 2.885          | 0.313    | 0.261     | 3.076 | 0.321    | 0.054     | 3.198      | 0.313    | 0.085     |
| IMMS-R      | 2.448            | 0.253    | 0.355     | 2.599 | 0.394    | 0.178     | 2.292      | 0.485    | 0.402     | 2.406          | 0.262    | 0.317     | 2.500 | 0.393    | 0.350     | 2.599      | 0.466    | 0.102     |
| IMMS-C      | 2.500            | 0.388    | 0.872     | 2.536 | 0.478    | 0.486     | 2.431      | 0.362    | 0.464     | 3.052          | 0.351    | 0.313     | 3.069 | 0.346    | 0.063     | 3.151      | 0.326    | 0.303     |
| IMMS-S      | 3.117            | 0.276    | 0.100     | 3.342 | 0.309    | 0.163     | 3.056      | 0.399    | 0.480     | 2.933          | 0.365    | 0.831     | 3.022 | 0.349    | 0.242     | 3.117      | 0.276    | 0.228     |
| SOCS        | 2.939            | 0.233    | 0.961     | 3.069 | 0.318    | 0.629     | 3.004      | 0.276    | 0.666     | 3.006          | 0.318    | 0.545     | 3.215 | 0.354    | 0.582     | 3.136      | 0.240    | 0.146     |
| CLSS_CM     | 2.938            | 0.269    | 0.093     | 3.078 | 0.241    | 0.581     | 3.000      | 0.300    | 0.105     | 3.229          | 0.310    | 0.271     | 3.201 | 0.291    | 0.498     | 3.089      | 0.426    | 0.604     |

GEQ\_SPM: EM - Empathy; NF - Negative Feelings; BI - Behavioral Involvement.

IMMS: A - Attention; R - Relevance; C - Confidence; S - Satisfaction.

E-E — Extrovert-Extrovert; I-I — Introvert-Introvert.

extrovert-extrovert pairs ( $\mu = 2.843$ ). Familiar pairs consistently reported higher confidence (IMMS-C) regardless of trait matching. The assumptions of normality (Shapiro-Wilk test) and homogeneity of variance (Levene's test, all  $p > .05$ ) were satisfied, validating the parametric tests.

To assess the significance of these trends, two-way ANOVAs were conducted. Table 7 shows the ANOVA results which revealed a significant main effect of familiarity on multiple measures: GEQ\_SPM-EM, GEQ\_SPM-BI, IMMS-A, IMMS-C, IMMS-S and CLSS\_CM ( $p < .05$ ). In contrast, personality trait composition alone did not produce a significant main effect on any measure. Significant interaction effects between familiarity and personality trait were found for GEQ\_SPM-BI, IMMS-A, and IMMS-S ( $p < .05$ ). This indicates that the influence of personality matching is contingent upon the level of pre-existing familiarity between teammates.

### 5.2.2. Post-hoc analysis of interaction effects

Figs. 5 and 6 present the results of Fisher's LSD post-hoc tests conducted to interpret the significant interactions. For GEQ\_SPM-BI and IMMS-A, a clear pattern is shown: familiarity significantly decreased scores for matched pairs (extrovert-extrovert and introvert-introvert) but had no effect on mismatched pairs. For example, unfamiliar extrovert-extrovert pairs reported significantly higher GEQ\_SPM-BI than familiar extrovert-extrovert pairs ( $\Delta = -.442$ ,  $p < .001$ ).

In addition, among familiar pairs, the matched pairs (both extrovert-extrovert and introvert-introvert) reported significantly higher GEQ\_SPM-BI and IMMS-A than the mismatched pairs. This discipline-based advantage was not present among unfamiliar pairs. The pattern for

IMMS-S differed. Only extrovert-extrovert pairs showed a significant familiarity effect, with unfamiliar pairs reporting higher satisfaction than familiar ones ( $\Delta = -.319$ ,  $p = .002$ ). Among unfamiliar pairs only, extrovert-extrovert pairs also reported higher IMMS-S than mismatched pairs. In summary, the results demonstrate that the impact of the alignment of personality traits is conditional on familiarity, although familiarity generally increased confidence, its effects on behavioral involvement, attention, and satisfaction were moderated by whether the pairs were personality-matched, with mismatched pairs generally insensitive to familiarity effects. Table A.7 reports the complete detailed results.

### 5.3. Observations analysis

The inter-rater reliability for the nine observation codes (A1-A9) was excellent. Table 8 shows a two-way mixed-effects intraclass correlation coefficient (ICC) (Shrout & Fleiss, 1979) that produced single measure ICC(3,1) values of 0.78 to 0.90 and average measure ICC(3,k) values of 0.87 to 0.95, indicating good-to-excellent consistency (Koo & Li, 2016). Subsequent analyses used the average scores of both raters.

#### 5.3.1. Descriptive statistical results

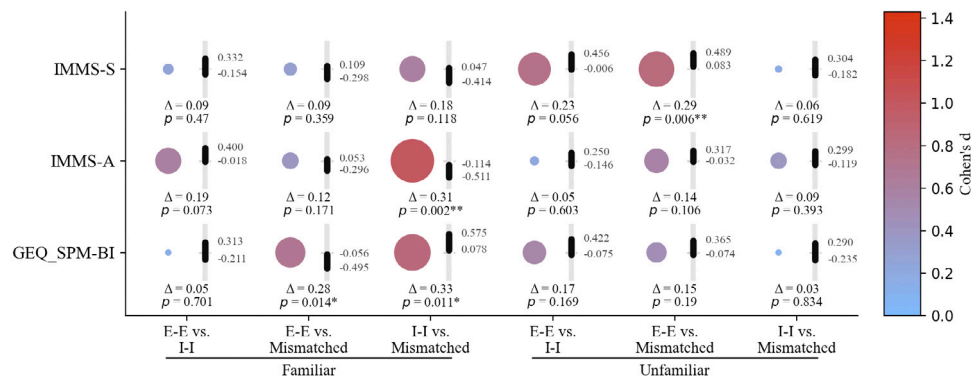
Tables 9 and 10 report descriptive statistics indicating a uniform improvement from Evaluation 1 ( $\mu_{E1}$ ) to Evaluation 3 ( $\mu_{E3}$ ) under all conditions, as shown by positive mean differences ( $\Delta$ ). Unfamiliar pairs generally started with lower  $\mu_{E1}$  scores than familiar pairs, although unfamiliar inter-faculty pairs showed similar or slightly higher initial performance in A5 (division of tasks) and A7 (technical coordination).

**Table 7**

Two-way ANOVA analysis of familiarity (FM), personality trait (PT), and their interactions (FM  $\times$  PT). Only measurements with statistically significant differences are shown and the detailed results are presented in Table A.13.

| Measurement | Source         | Type III sum of squares | df | Mean square | F      | p         | $\eta^2 p$ |
|-------------|----------------|-------------------------|----|-------------|--------|-----------|------------|
| GEQ_SPM-EM  | FM             | 0.497                   | 1  | 0.497       | 4.726  | 0.032*    | 0.044      |
| GEQ_SPM-BI  | FM             | 1.695                   | 1  | 1.695       | 13.466 | <0.001*** | 0.117      |
|             | FM $\times$ PT | 0.949                   | 2  | 0.474       | 3.768  | 0.026*    | 0.069      |
| IMMS-A      | FM             | 1.046                   | 1  | 1.046       | 13.106 | <0.001*** | 0.114      |
|             | FM $\times$ PT | 0.699                   | 2  | 0.349       | 4.379  | 0.015*    | 0.079      |
| IMMS-C      | FM             | 9.028                   | 1  | 9.028       | 61.471 | <0.001*** | 0.376      |
| IMMS-S      | FM             | 0.540                   | 1  | 0.540       | 4.995  | 0.028*    | 0.047      |
|             | FM $\times$ PT | 0.758                   | 2  | 0.379       | 3.507  | 0.034*    | 0.064      |
| CLSS_CM     | FM             | 0.702                   | 1  | 0.702       | 6.927  | 0.010*    | 0.064      |

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .



**Fig. 5.** Fisher's LSD post-hoc pairwise comparisons of GEQ\_SPM-BI, IMMS-A, and IMMS-S between for the main effects between familiarity and personality trait.  $\Delta$  represents the mean difference between groups;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Table 8**

Inter-rater reliability analysis for observational codes using two-way mixed-effects intraclass correlation coefficients for both single and average measures, reflecting the level of agreement between raters. F is the test statistic with associated degrees of freedom (df1, df2);  $p$  denotes statistical significance.

| Code | ICC type         | Intraclass correlation | 95% CI |       | F test with true value 0 |     |     |           |
|------|------------------|------------------------|--------|-------|--------------------------|-----|-----|-----------|
|      |                  |                        | LL     | UL    | Value                    | df1 | df2 | p         |
| A1   | Single measures  | 0.902                  | 0.859  | 0.932 | 19.335                   | 107 | 107 | <0.001*** |
|      | Average measures | 0.948                  | 0.924  | 0.965 | 19.335                   | 107 | 107 | <0.001*** |
| A2   | Single measures  | 0.821                  | 0.748  | 0.874 | 10.142                   | 107 | 107 | <0.001*** |
|      | Average measures | 0.901                  | 0.856  | 0.933 | 10.142                   | 107 | 107 | <0.001*** |
| A3   | Single measures  | 0.810                  | 0.734  | 0.866 | 9.536                    | 107 | 107 | <0.001*** |
|      | Average measures | 0.895                  | 0.847  | 0.928 | 9.536                    | 107 | 107 | <0.001*** |
| A4   | Single measures  | 0.850                  | 0.788  | 0.895 | 12.364                   | 107 | 107 | <0.001*** |
|      | Average measures | 0.919                  | 0.882  | 0.945 | 12.364                   | 107 | 107 | <0.001*** |
| A5   | Single measures  | 0.837                  | 0.770  | 0.885 | 11.243                   | 107 | 107 | <0.001*** |
|      | Average measures | 0.911                  | 0.870  | 0.939 | 11.243                   | 107 | 107 | <0.001*** |
| A6   | Single measures  | 0.837                  | 0.770  | 0.885 | 11.243                   | 107 | 107 | <0.001*** |
|      | Average measures | 0.911                  | 0.870  | 0.939 | 11.243                   | 107 | 107 | <0.001*** |
| A7   | Single measures  | 0.836                  | 0.769  | 0.885 | 11.185                   | 107 | 107 | <0.001*** |
|      | Average measures | 0.911                  | 0.869  | 0.939 | 11.185                   | 107 | 107 | <0.001*** |
| A8   | Single measures  | 0.776                  | 0.688  | 0.841 | 7.924                    | 107 | 107 | <0.001*** |
|      | Average measures | 0.874                  | 0.815  | 0.914 | 7.924                    | 107 | 107 | <0.001*** |
| A9   | Single measures  | 0.783                  | 0.698  | 0.847 | 8.220                    | 107 | 107 | <0.001*** |
|      | Average measures | 0.878                  | 0.822  | 0.917 | 8.220                    | 107 | 107 | <0.001*** |

**Table 9**

Descriptive statistics for observations, including mean scores of two researchers in Evaluation 1 ( $\mu_{E1}$ ) and Evaluation 3 ( $\mu_{E3}$ ), as well as mean differences ( $\Delta = \mu_{E3} - \mu_{E1}$ ), by familiarity and academic discipline.

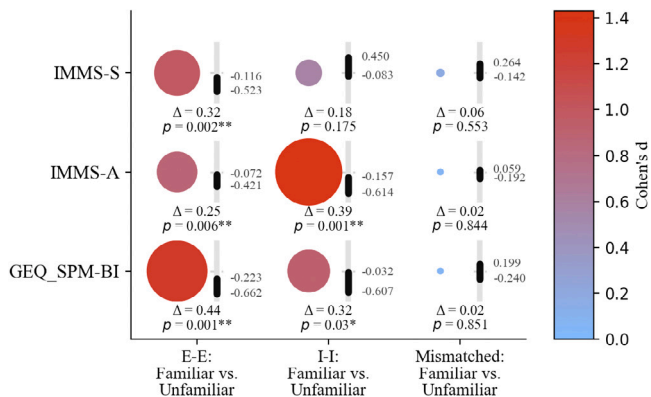
| Code | Unfamiliar pairs |            |          |               |            |          | Familiar pairs |            |          |               |            |          |
|------|------------------|------------|----------|---------------|------------|----------|----------------|------------|----------|---------------|------------|----------|
|      | Intra-Faculty    |            |          | Inter-Faculty |            |          | Intra-Faculty  |            |          | Inter-Faculty |            |          |
|      | $\mu_{E1}$       | $\mu_{E3}$ | $\Delta$ | $\mu_{E1}$    | $\mu_{E3}$ | $\Delta$ | $\mu_{E1}$     | $\mu_{E3}$ | $\Delta$ | $\mu_{E1}$    | $\mu_{E3}$ | $\Delta$ |
| A1   | 1.81             | 3.46       | 1.65     | 2.00          | 3.36       | 1.36     | 2.25           | 3.50       | 1.25     | 2.40          | 3.50       | 1.10     |
| A2   | 2.00             | 3.23       | 1.23     | 2.32          | 3.32       | 1.00     | 2.17           | 3.58       | 1.42     | 2.47          | 3.57       | 1.10     |
| A3   | 2.19             | 3.35       | 1.15     | 2.26          | 3.39       | 1.04     | 2.17           | 3.67       | 1.50     | 2.47          | 3.50       | 1.03     |
| A4   | 2.23             | 3.31       | 1.08     | 2.14          | 3.32       | 1.18     | 2.71           | 3.54       | 0.83     | 2.83          | 3.67       | 0.83     |
| A5   | 2.08             | 3.15       | 1.08     | 2.39          | 3.32       | 0.93     | 2.21           | 3.46       | 1.25     | 2.27          | 3.57       | 1.30     |
| A6   | 2.15             | 2.85       | 0.69     | 2.14          | 2.39       | 0.25     | 2.58           | 2.79       | 0.21     | 2.60          | 2.70       | 0.10     |
| A7   | 1.54             | 2.85       | 1.31     | 1.64          | 2.89       | 1.25     | 2.25           | 3.54       | 1.29     | 2.23          | 3.37       | 1.13     |
| A8   | 2.08             | 3.08       | 1.00     | 2.07          | 3.21       | 1.14     | 2.75           | 3.50       | 0.75     | 2.90          | 3.50       | 0.60     |
| A9   | 2.15             | 3.35       | 1.19     | 2.25          | 3.43       | 1.18     | 2.50           | 3.54       | 1.05     | 2.60          | 3.63       | 1.03     |

**Table 10**

Descriptive statistics for observations, including mean scores of two researchers in Evaluation 1 ( $\mu_{E1}$ ) and Evaluation 3 ( $\mu_{E3}$ ), as well as mean differences ( $\Delta = \mu_{E3} - \mu_{E1}$ ), by familiarity and personality trait.

| Code | Unfamiliar pairs |            |          |            |            |          |            |            |          | Familiar pairs |            |          |            |            |          |            |            |          |
|------|------------------|------------|----------|------------|------------|----------|------------|------------|----------|----------------|------------|----------|------------|------------|----------|------------|------------|----------|
|      | I-I              |            |          | E-E        |            |          | Mismatched |            |          | I-I            |            |          | E-E        |            |          | Mismatched |            |          |
|      | $\mu_{E1}$       | $\mu_{E3}$ | $\Delta$ | $\mu_{E1}$ | $\mu_{E3}$ | $\Delta$ | $\mu_{E1}$ | $\mu_{E3}$ | $\Delta$ | $\mu_{E1}$     | $\mu_{E3}$ | $\Delta$ | $\mu_{E1}$ | $\mu_{E3}$ | $\Delta$ | $\mu_{E1}$ | $\mu_{E3}$ | $\Delta$ |
| A1   | 1.25             | 3.08       | 1.83     | 1.96       | 3.54       | 1.58     | 2.31       | 3.44       | 1.13     | 2.08           | 3.25       | 1.17     | 2.33       | 3.67       | 1.33     | 2.46       | 3.50       | 1.04     |
| A2   | 1.50             | 3.17       | 1.67     | 2.35       | 3.27       | 0.92     | 2.38       | 3.38       | 1.00     | 1.92           | 3.50       | 1.58     | 2.39       | 3.61       | 1.22     | 2.50       | 3.58       | 1.08     |
| A3   | 1.08             | 2.33       | 1.25     | 2.77       | 3.81       | 1.04     | 2.38       | 3.44       | 1.06     | 1.92           | 3.33       | 1.42     | 2.61       | 3.83       | 1.22     | 2.33       | 3.50       | 1.17     |
| A4   | 1.67             | 3.17       | 1.50     | 2.23       | 3.31       | 1.08     | 2.50       | 3.44       | 0.94     | 2.25           | 3.17       | 0.92     | 2.89       | 3.89       | 1.00     | 2.96       | 3.63       | 0.67     |
| A5   | 1.17             | 2.33       | 1.17     | 2.81       | 3.54       | 0.73     | 2.13       | 3.44       | 1.31     | 1.75           | 3.17       | 1.42     | 2.56       | 3.78       | 1.22     | 2.25       | 3.50       | 1.25     |
| A6   | 2.08             | 2.92       | 0.83     | 1.85       | 2.35       | 0.50     | 2.69       | 2.81       | 0.13     | 2.17           | 2.42       | 0.25     | 2.67       | 2.72       | 0.06     | 2.75       | 2.92       | 0.17     |
| A7   | 0.92             | 2.33       | 1.42     | 1.42       | 2.65       | 1.23     | 2.38       | 3.63       | 1.25     | 1.67           | 3.33       | 1.67     | 2.56       | 3.50       | 0.94     | 2.29       | 3.46       | 1.17     |
| A8   | 1.75             | 2.92       | 1.17     | 1.96       | 3.04       | 1.08     | 2.50       | 3.50       | 1.00     | 2.42           | 3.25       | 0.83     | 2.67       | 3.61       | 0.94     | 3.17       | 3.54       | 0.38     |
| A9   | 1.33             | 2.92       | 1.58     | 2.35       | 3.54       | 1.19     | 2.63       | 3.50       | 0.88     | 2.00           | 3.25       | 1.25     | 3.00       | 3.78       | 0.78     | 2.50       | 3.63       | 1.13     |

E-E — Extrovert–Extrovert; I-I — Introvert–Introvert.



**Fig. 6.** Fisher's LSD post-hoc pairwise comparisons of GEQ-SPM-BI, IMMS-A, and IMMS-S for the interactions between familiarity and personality trait.  $\Delta$  represents the mean difference between groups;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

By Evaluation 3, unfamiliar inter-faculty pairs matched or exceeded other groups in codes like A1 (maintain mutual understanding), A3 (information set), A4 (reaching consensus) and A9 (orientation of individual tasks).

The magnitude of the gain ( $\Delta$ ) varied by condition. Unfamiliar pairs, regardless of academic discipline, showed substantial improvements in mutual understanding (A1), dialog management (A2), and technical coordination (A7). In contrast, code A6 (time management) showed consistently minimal improvement in all conditions, suggesting that the intervention had little effect on this specific behavior. When considering personality, unfamiliar introvert–introvert pairs exhibited the

largest gains in A1, A2, and A9, while familiar introvert–introvert pairs improved the most in A2, A5 and A7. These descriptive patterns suggested complex interactions between familiarity, academic discipline, and personality trait, which requires formal statistical testing.

### 5.3.2. Interaction effects analysis

Before analysis, the Shapiro–Wilk tests indicated significant deviations from normality ( $p < 0.05$ ) for the variables of interest. Therefore, generalized linear models (GLMs) were used and Wald  $\chi^2$  statistics with partial  $\eta_p^2$  are reported. Table 11 presents the key GLM results. For the model examining familiarity and academic discipline, significant main effects of familiarity were found for A1 ( $p = .010$ ), A4 ( $p = .024$ ), A5 ( $p = .015$ ), A6 ( $p < .001$ ) and A8 ( $p = .001$ ). The academic discipline showed significant main effects for A2 ( $p = .028$ ), A3 ( $p = .022$ ), and A6 ( $p = .003$ ). The interaction between familiarity and academic discipline was not significant for any code.

Significant effects were also found for the model that examined familiarity and personality traits. Familiarity had significant main effects for A1, A4, A5, A6, and A8. The personality composition had significant main effects for A1, A2, A4, A5, A6, A7, and A8. Specifically, a significant interaction between familiarity and personality traits was found for A6 ( $p = .018$ ), A8 ( $p = .021$ ) and A9 ( $p = .021$ ), indicating that the effect of personality on these behaviors depends on the level of familiarity between teammates. These observational results help explain the statistical effects: the interaction suggests that the coordination challenges in time management (A6) and task orientation (A9) within mismatched personality pairs are particularly marked in unfamiliar settings, a dynamic not captured by main effects alone.

### 5.3.3. Post-hoc pairwise comparisons

Figs. 7, 8, and 9 summarize the post-hoc estimated marginal means (EMM) for the significant main and interaction effects involving the personality trait (see Tables A.8 and A.9 for full results). For main effects, mismatched pairs scored significantly lower in A1 (mutual

**Table 11**

Generalized linear model (GLM) analysis of familiarity (FM) and academic discipline (AD), including their interaction (FM  $\times$  AD), as well as familiarity (FM) and personality trait (PT), including their interaction (FM  $\times$  PT). Only measurements with statistically significant differences are shown and the detailed results are presented in Table A.10.

| Code | Source         | Wald Chi-Square | df | p         | $\eta^2 p$ | Source         | Wald Chi-Square | df | p         | $\eta^2 p$ |
|------|----------------|-----------------|----|-----------|------------|----------------|-----------------|----|-----------|------------|
| A1   | FM             | 6.657           | 1  | 0.010*    | 0.058      | FM             | 6.584           | 1  | 0.010*    | 0.057      |
|      | AD             | 3.041           | 1  | 0.081     | 0.027      | PT             | 8.993           | 2  | 0.011*    | 0.077      |
| A2   | AD             | 4.806           | 1  | 0.028*    | 0.043      | PT             | 15.974          | 2  | <0.001*** | 0.129      |
| A3   | AD             | 5.233           | 1  | 0.022*    | 0.046      | PT             | 1.800           | 2  | 0.407     | 0.016      |
| A4   | FM             | 5.076           | 1  | 0.024*    | 0.045      | FM             | 5.575           | 1  | 0.018*    | 0.049      |
|      | AD             | 0.151           | 1  | 0.697     | 0.001      | PT             | 6.100           | 2  | 0.047*    | 0.053      |
| A5   | FM             | 5.880           | 1  | 0.015*    | 0.052      | FM             | 4.167           | 1  | 0.041*    | 0.037      |
|      | AD             | 0.192           | 1  | 0.661     | 0.002      | PT             | 7.959           | 2  | 0.019*    | 0.069      |
| A6   | FM             | 11.435          | 1  | <0.001*** | 0.096      | FM             | 11.776          | 1  | <0.001*** | 0.098      |
|      | AD             | 8.626           | 1  | 0.003**   | 0.074      | PT             | 10.270          | 2  | 0.006**   | 0.087      |
|      | FM $\times$ AD | 3.173           | 1  | 0.075     | 0.029      | FM $\times$ PT | 8.060           | 2  | 0.018*    | 0.069      |
| A7   | AD             | 0.716           | 1  | 0.398     | 0.007      | PT             | 7.791           | 2  | 0.019*    | 0.069      |
| A8   | FM             | 10.552          | 1  | 0.001**   | 0.089      | FM             | 8.861           | 1  | 0.003**   | 0.076      |
|      | AD             | 0.001           | 1  | 0.977     | 0.000      | PT             | 6.751           | 2  | 0.034*    | 0.059      |
| A9   | AD             | 0.009           | 1  | 0.925     | 0.000      | PT             | 10.400          | 2  | 0.006**   | 0.088      |
|      | FM $\times$ AD | 0.001           | 1  | 0.982     | 0.000      | FM $\times$ PT | 7.726           | 2  | 0.021*    | 0.067      |

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

understanding) than in both matched personality groups. In contrast, extrovert–extrovert pairs scored lower in A5 (division of tasks) and A9 (task orientation) than both introvert–introvert and mismatched pairs, indicating potential limitations in systematic task execution despite strengths in verbal exchange.

The interaction between familiarity and personality traits for A6 and A9 revealed significant patterns. For time management (A6), both extrovert–extrovert and introvert–introvert pairs improved significantly more with unfamiliar teammates, while mismatched pairs were not affected by familiarity and consistently showed the poorest coordination in unfamiliar settings. This observational finding explains the statistical interaction: the benefit of an unfamiliar teammate for matched personalities does not extend to mismatched pairs, who struggle with time coordination regardless of familiarity. For task orientation (A9), unfamiliar extrovert–extrovert pairs improved significantly over familiar ones. Furthermore, the effect of personality composition depended on familiarity: in familiar settings, extrovert–extrovert pairs outperformed introvert–introvert pairs; in unfamiliar settings, introvert–introvert pairs outperformed both other groups.

In summary, the results indicate that the influence of personality on collaborative social interaction is moderated by familiarity. Even though matched personality pairs show specific advantages, these are often amplified in unfamiliar settings, prompting more structured collaboration. However, mismatched pairs are less influenced by familiarity and exhibit persistent coordination difficulties in time management and task orientation, particularly when unfamiliar. Thus, observational data go beyond supporting quantitative effects to explain the underlying mechanism: familiarity mitigates coordination deficits for matched personalities but fails to do so for mismatched pairs, highlighting an inherent coordination challenge in mismatched teams.

#### 5.4. Interviews analysis

Table 12 presents the thematic analysis of participant interviews, indicating largely positive experiences in all game dimensions and the cooperative interaction. For individual experience, 89.8% of responses reflected positive emotions, high engagement, and a strong sense of personal achievement, indicating that cooperative IVR tasks effectively evoked enjoyment and intrinsic motivation. For team collaboration, 88.0% of the responses highlighted positive experiences in comfort, trust, and effective teamwork. Finally, 86.1% of the responses indicated

positive reflections on skill improvement and willingness for future participation. Negative feedback was infrequent and mainly concerned initial anxiety with unfamiliar teammates, minor communication barriers, physical discomfort, and technical issues.

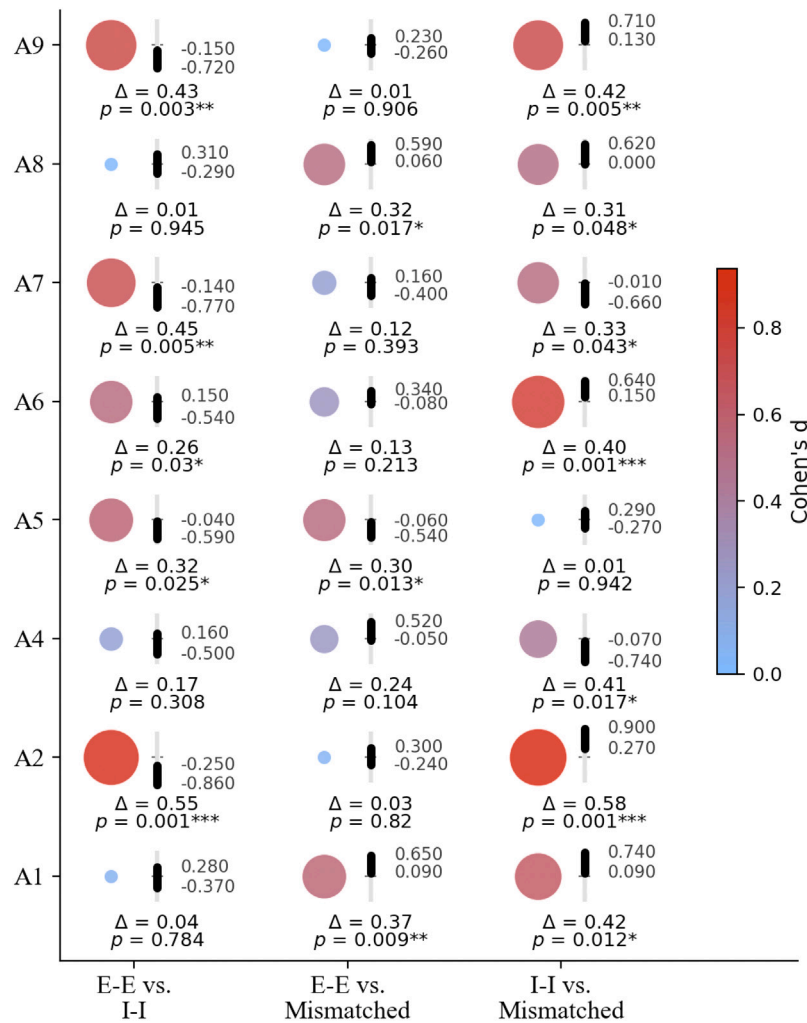
Beyond these distributions, the analysis yielded explanatory themes that elucidate the mechanisms behind key quantitative outcomes. First, the described initial “warming up” phase in unfamiliar inter-faculty pairs, characterized by extra explanatory effort due to disciplinary differences, provides a process explanation for the statistical pattern in which these groups demonstrated high behavioral involvement but comparatively lower scores on measures of perceived social interdependence. Second, participants in familiar intra-faculty pairs reported efficient coordination based on pre-existing trust, which aligns with their higher satisfaction scores. However, their frequent reliance on implicit routines with limited explicit discussion introduces a subtle tension, suggesting that high satisfaction may sometimes correlate with less verbalized co-construction of knowledge. Third, within several unfamiliar and extrovert–extrovert pairs, heterogeneity in knowledge and communication style was often framed as a constructive learning opportunity. This active involvement in questioning and joint problem-solving helps explain the significant gains in procedural knowledge observed in these conditions, positioning communication challenges as catalysts for deeper cognitive engagement.

Together, these themes position qualitative findings as explanatory, indicating that identical quantitative outcomes may stem from different social mechanisms, and that cooperative IVR tasks concurrently reveal communication challenges and opportunities for interpersonal learning.

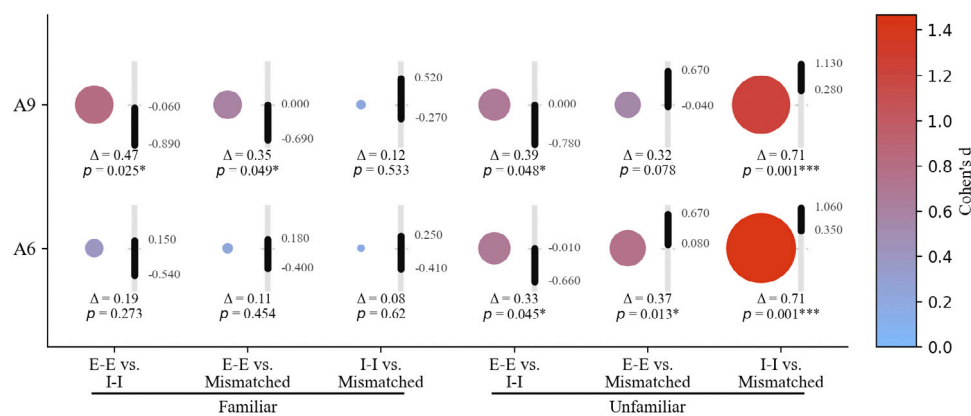
## 6. Discussion

**RQ1: How does pairing unfamiliar students from different faculties impact social interaction within a two-player cooperative IVR game?**

The analysis led to the rejection of  $H_{01}$ , revealing a statistically significant interaction effect between interpersonal familiarity and academic discipline on social interaction. Unfamiliar inter-faculty pairs demonstrated greater behavioral involvement, while familiar intra-faculty pairs exhibited stronger perceived social interdependence. A main effect of discipline was also found, with inter-faculty pairs rating the task as more relevant. In particular, no condition showed an increased negative affect, indicating that increased engagement did not come at the cost of discomfort.



**Fig. 7.** EMM post-hoc pairwise comparisons of observational codes by personality trait (PT).  $\Delta$  represents the mean difference between groups;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



**Fig. 8.** EMM post-hoc pairwise comparisons of observational codes A6 and A9 for the main effects between familiarity and personality trait.  $\Delta$  represents the mean difference between groups;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

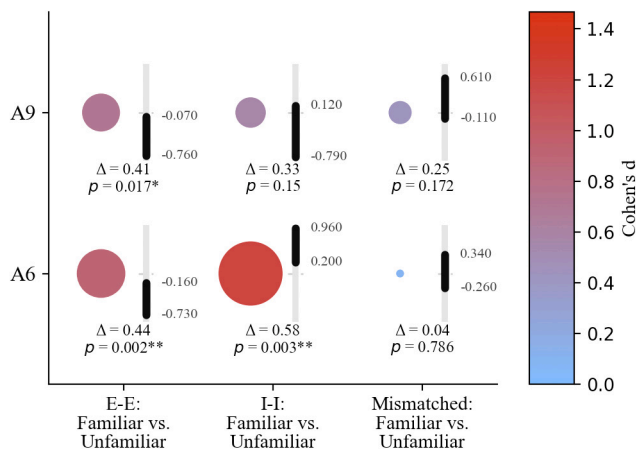
These results suggest two distinct pathways of interaction. For unfamiliar inter-faculty pairs, higher involvement appeared to be associated with the need for clarification and knowledge integration. Disciplinary differences necessitated intensive “translation, interpretation, and alignment” (Järvelä & Hadwin, 2013), manifesting itself as

frequent joint monitoring and behavioral participation, consistent with the greatest gains in observation codes, the result align with Horn et al. (2023), they demonstrated that interdisciplinary collaboration should address disciplinary differences and achieve mutual understanding and knowledge integration through continuous dialog and negotiation. In

**Table 12**

Thematic analysis of participants interviews responses showing the prevalence of positive over negative experiences.

| Part No. | Experience | Aspect                                               | Count | Total (%)  |
|----------|------------|------------------------------------------------------|-------|------------|
| Part 1   | Positive   | - Positive emotions                                  | 56    | 97 (89.8%) |
|          |            | - Engagement and immersion                           | 9     |            |
|          |            | - Achievement and Self-Worth (Individual Experience) | 32    |            |
|          | Negative   | - Emotional tension and anxiety                      | 7     | 11 (10.2%) |
|          |            | - Frustration and discouragement                     | 4     |            |
| Part 2   | Positive   | - Comfort and trust                                  | 24    | 95 (88.0%) |
|          |            | - Effective teamwork                                 | 28    |            |
|          |            | - Positive interaction                               | 22    |            |
|          |            | - Successful outcomes (Team Collaboration)           | 21    |            |
|          | Negative   | - Nervousness (Initial Stage of Teamwork)            | 6     | 13 (12.0%) |
|          |            | - Communication barriers                             | 7     |            |
| Part 3   | Positive   | - Improved Skills                                    | 20    | 93 (86.1%) |
|          |            | - Personal growth                                    | 19    |            |
|          |            | - Future willingness                                 | 23    |            |
|          |            | - General positive reflection                        | 31    |            |
|          | Negative   | - Physical discomfort                                | 13    | 15 (13.9%) |
|          |            | - Technical issues                                   | 2     |            |



**Fig. 9.** EMM post-hoc pairwise comparisons of observational codes A6 and A9 for the interactions between familiarity and personality trait.  $\Delta$  represents the mean difference between groups;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

contrast, familiar intra-faculty pairs utilized their shared background for efficient coordination, resulting in clearer task division, lower coordination costs, and higher social interdependence ratings, consistent with Nawata et al. (2020) showing that team familiarity and shared experience foster transactive memory, implicit coordination, and reduced communication overhead.

Motivationally, interdisciplinary settings were associated with higher perceived task relevance, while familiarity increased perceived competence and confidence. This aligns with SDT (Ryan, 1982), where contexts that support autonomy, competence, and relatedness promote engagement. The data from the interviews clarify these mechanisms. Participant U12-PB (unfamiliar, inter-faculty) noted:

*"I wanted to explain to my teammate the knowledge that I was familiar with, but I found that he couldn't understand... During the game, it took time to continue explaining and confirming."*

In contrast, participant F02-PA (familiar, intra-faculty) stated:

*"We didn't communicate much during the game, basically only at the beginning... we had a clear division of labor and did not need much discussion."*

The observation that disciplinary diversity co-occurred with greater behavioral engagement without eliciting negative emotions suggests that, when tasks integrate informational interdependence, heterogeneity functions as a cognitive investment rather than a social burden (Van Dijk et al., 2012).

This supports existing work in which heterogeneity can improve participation through elaboration (Homan et al., 2020; Mesmer-Magnus & DeChurch, 2009), while shared perspectives within familiar teams reduce coordination costs (Huckman et al., 2009; Pasarakonda et al., 2025). The observation that disciplinary diversity co-occurred with greater behavioral engagement without negative emotions suggests that, when tasks integrate informational interdependence, heterogeneity functions as a cognitive investment rather than a social burden (Van Dijk et al., 2012). Importantly, the observed disciplinary effects arise from a combination of the diversity of the inherent academic background and the specific informational asymmetries designed into the escape-room task, rather than from fixed disciplinary differences alone. The participant U17-PB illustrated this:

*"When I learn something more than I know, I think it is interesting and I am willing to talk more... the process is fun."*

Even though shared mental models improve coordination efficiency (Mohammed et al., 2010), these findings reveal that alignment and divergence function as complementary mechanisms. The convergence of mental models optimizes execution, while cognitive divergence maintains broader engagement. Although unfamiliarity can reduce communication (Edmondson, 1999; Robinson & Held, 2025), the high interdependence of tasks in this study co-occurred with sustained interaction and may have reduced uncertainty (Kozlowski & Ilgen, 2006), consistent with strong gains in mutual understanding for unfamiliar pairs (Lisak et al., 2022).

Both observational and interview data showed a consistent temporal pattern. The initial high levels of communication and alignment decreased as the game progressed. The ongoing interaction reduced initial communication costs, especially for unfamiliar pairs, who focused intensively on synchronizing efforts before shifting to implicit coordination. As U12-PA stated:

*"It was difficult to communicate at the beginning... but by the end we had developed an implicit understanding."*

Qualitatively, unfamiliar inter-faculty pairs converted early, effortful coordination into high behavioral involvement, while familiar intra-faculty pairs relied on pre-established routines to sustain social interdependence.

In sum, a significant interaction between familiarity and discipline shapes cooperative dynamics. For educational goals that emphasize exploration and knowledge integration, educators designing short puzzle-based IVR activities may consider forming unfamiliar inter-faculty pairs and integrating information asymmetries in tasks, as this configuration was associated with greater communication and behavioral involvement in the present study. When objectives require tightly coupled execution, familiar intra-faculty pairs are preferable. Integrating attention supports (e.g., stage objectives, temporal cues) in the IVR interface can address time management. The alignment of the team composition with the task structure provides a concrete design guideline for configuring cooperative IVR courses to serve different pedagogical goals.

**RQ2: How does pairing unfamiliar students with mismatched personality traits impact social interaction within a two-player cooperative IVR game?**

The analysis also rejected  $H_{02}$ , indicating a significant interaction between familiarity and personality composition. Familiar pairs generally outperformed unfamiliar pairs in measures of behavioral involvement, confidence, and collaborative satisfaction, which can reflect stronger implicit coordination and collective efficiency among familiar pairs. However, this was qualified by a significant interaction with personality. This pattern aligns with routine-based collaboration: familiar pairs can rely on shared mental models, reducing the need for explicit monitoring and potentially leading to lower subjective reports of “attention” even when performance is high (Järvelä & Hadwin, 2013).

Mismatched personality pairs showed relative insensitivity to familiarity in terms of behavioral involvement. A central discrepancy was identified in reported subjective experience: unfamiliar extrovert-extrovert pairs reported the highest satisfaction and attention, while unfamiliar mismatched pairs reported the lowest. This aligns with the ability of extroverts’ to generate positive affect and mitigate the challenges of unfamiliar collaboration (Laakasuo et al., 2020), a trait that is effective in virtual settings (Mueller et al., 2025; Zhu et al., 2024). The persistently low satisfaction in unfamiliar mismatched pairs highlights that similarity in personality and familiarity is a prominent factor in perceived fit (Sened et al., 2025). Participant F21-PA (familiar, matched) noted:

*“My teammates and I are already good friends... we don’t need to talk much.”*

In contrast, participant U30-PA (unfamiliar, matched) stated:

*“Even though my teammate and I didn’t know each other, we had friendly discussions... I am very happy with how we worked together.”*

Consequently, for short puzzle-based IVR activities aimed at increasing motivational satisfaction and exploratory engagement, instructors can consider pairing unfamiliar extrovert-extrovert students, as this configuration was associated with the highest reported satisfaction in the present dataset. When the goal is stable coordination and collective regulation, familiar pairs (regardless of personality matching) are reliable. Unfamiliar mismatched pairs, which reported the lowest satisfaction, should therefore be accompanied by additional support, such as structured role assignments or explicit collaboration scripts, to sustain engagement. These findings highlight the importance of integrating familiarity and personality into collaborative IVR design through explicit choices about team formation and scaffolding.

## 6.1. Limitations and future work

The contributions of this study must be interpreted in light of its limitations, which also indicate promising directions for future work. These results are based on an asymmetric escape-room-like IVR task with a homogeneous sample of university students. Consequently, the results are most directly transferable to similar cooperative, puzzle-based immersive learning environments that require high interdependence and information exchange among tertiary-level learners. Given the specific IVR escape-room context with asymmetrical tasks, caution is needed when generalizing the findings. This requires caution when extrapolating to substantially different populations (e.g., younger learners, working professionals), alternative collaborative structures (e.g., low-interdependence or fully symmetric tasks), or other disciplinary settings. Future studies should examine whether the identified interactions between familiarity, discipline, and personality generalize to wider contexts and diverse IVR task designs, thus clarifying the limits of our conclusions.

Second, the operationalization of personality was deliberately narrow and focused solely on the extraversion-introversion dimension. Even though this provided a clear analytical focus, it omits other traits that likely influence collaborative dynamics. For instance, conscientiousness might affect task coordination and persistence, while agreeableness could shape conflict management and mutual support, factors not captured here. Consequently, the findings on personality similarity are limited to this single-dimension trait; using a multi-dimension trait framework (e.g. the Big Five) would likely produce a more comprehensive and significant account of personality’s influence on social interaction in IVR.

Third, the finding that familiar pairs reported lower attention demands deeper theoretical engagement. The brief speculation points to potential routine-based collaboration and implicit coordination, where established pairs may rely on shared scripts and reduced explicit communication. Future work should empirically test this explanation by integrating process measures (e.g. communication analysis, dual-eye tracking) with established literature on team coordination to determine whether reduced attention reflects efficiency or a potential complacency effect. This requires a dedicated qualitative and quantitative investigation.

Lastly, although self-report measures were analyzed at the individual level to preserve each participant’s subjective evaluation of the collaborative IVR experience, the responses were collected within pairs and thus may not have been fully independent; moreover, the study did not explicitly model interdependence between partner responses.

Future research should consider using analytical approaches that account for non-independence between partners, such as multilevel models or other techniques that explicitly model interdependence. In addition, the sequential task order, typical of escape rooms, introduces potential practice effects; counterbalancing tasks would help mitigate this risk. Complementing self-reports with objective performance metrics (e.g. completion time, solution accuracy) would strengthen measurement, and longitudinal designs could assess the sustainability of observed effects on learning and social engagement over time.

## 7. Conclusions

This study investigated the complex interplay of interpersonal familiarity, academic discipline, and personality in shaping social engagement during a cooperative IVR learning task. Its primary strengths lie in its experimental examination of these multifaceted social factors within a single, high-fidelity, controlled cooperative IVR experiment, thus moving beyond simple main-effects models and prior approaches that have typically examine these factors in isolation to reveal critical interaction effects that define their contextual relevance.

Theoretically, the findings contribute a dual-pathway account to models of collaborative learning in immersive environments. This study

**Table A.1**

Demographics of participants in familiar pairs, categorized by academic discipline (three faculties) and personality trait (extrovert or introvert). Participants are labeled Participant A (PA) and Participant B (PB) within each pair (ID: FXX). Faculty abbreviations: Business (BS), Science & Engineering (SE), and Social Sciences (SS). A ✓denotes an intra-faculty pairing or matched personality trait.

| ID     | Age | Faculty | Personality | ID     | Age | Faculty | Personality | Intra-Faculty | Matched personality trait |
|--------|-----|---------|-------------|--------|-----|---------|-------------|---------------|---------------------------|
| F01-PA | 24  | BS      | Extrovert   | F01-PB | 23  | BS      | Extrovert   | ✓             | ✓                         |
| F02-PA | 21  | SE      | Introvert   | F02-PB | 22  | SE      | Introvert   | ✓             | ✓                         |
| F05-PA | 20  | SE      | Extrovert   | F05-PB | 22  | SE      | Introvert   | ✓             |                           |
| F07-PA | 21  | SE      | Extrovert   | F07-PB | 20  | BS      | Extrovert   |               | ✓                         |
| F08-PA | 22  | BS      | Introvert   | F08-PB | 24  | SE      | Extrovert   |               |                           |
| F09-PA | 21  | SE      | Introvert   | F09-PB | 21  | BS      | Extrovert   |               |                           |
| F10-PA | 20  | SS      | Introvert   | F10-PB | 22  | SE      | Extrovert   |               |                           |
| F11-PA | 24  | SE      | Extrovert   | F11-PB | 25  | SS      | Introvert   |               |                           |
| F13-PA | 20  | SS      | Introvert   | F13-PB | 25  | SS      | Extrovert   | ✓             |                           |
| F21-PA | 20  | SS      | Extrovert   | F21-PB | 19  | SS      | Extrovert   | ✓             | ✓                         |
| F22-PA | 21  | SS      | Extrovert   | F22-PB | 21  | SS      | Introvert   | ✓             |                           |
| F23-PA | 21  | SE      | Extrovert   | F23-PB | 22  | SE      | Extrovert   | ✓             | ✓                         |
| F24-PA | 21  | SE      | Introvert   | F24-PB | 21  | SS      | Introvert   |               |                           |
| F26-PA | 21  | SS      | Introvert   | F26-PB | 21  | SS      | Introvert   | ✓             | ✓                         |
| F27-PA | 20  | SE      | Introvert   | F27-PB | 21  | SS      | Extrovert   |               |                           |
| F32-PA | 21  | BS      | Extrovert   | F32-PB | 20  | SE      | Introvert   |               |                           |
| F33-PA | 22  | SE      | Introvert   | F33-PB | 23  | SE      | Introvert   | ✓             | ✓                         |
| F34-PA | 21  | BS      | Extrovert   | F34-PB | 21  | SE      | Extrovert   |               | ✓                         |
| F35-PA | 21  | BS      | Extrovert   | F35-PB | 21  | SE      | Introvert   |               |                           |
| F40-PA | 21  | BS      | Introvert   | F40-PB | 21  | BS      | Introvert   | ✓             | ✓                         |
| F41-PA | 23  | SS      | Extrovert   | F41-PB | 23  | BS      | Extrovert   |               | ✓                         |
| F47-PA | 24  | SE      | Extrovert   | F47-PB | 24  | SS      | Introvert   |               |                           |
| F48-PA | 23  | SE      | Extrovert   | F48-PB | 21  | SE      | Extrovert   | ✓             | ✓                         |
| F49-PA | 21  | SS      | Extrovert   | F49-PB | 20  | BS      | Extrovert   |               | ✓                         |
| F50-PA | 19  | SE      | Extrovert   | F50-PB | 22  | SS      | Extrovert   |               | ✓                         |
| F51-PA | 23  | SE      | Introvert   | F51-PB | 23  | BS      | Introvert   |               | ✓                         |
| F52-PA | 22  | SS      | Introvert   | F52-PB | 22  | SS      | Extrovert   | ✓             |                           |

reveals two distinct yet complementary social routes: one in which unfamiliar, inter-disciplinary pairs achieve engagement through explicit explanation and knowledge translation, and another in which familiar, intra-disciplinary pairs utilize shared practices and implicit coordination. This framework reframes the homophily–heterogeneity debate by demonstrating that their advantages depend on the collaboration stage, exploration or execution, and on transitions between explicit and implicit forms of social regulation. The counterintuitive finding of lower attention in familiar pairs further suggests the need to integrate theories of routine-based teamwork into immersive learning literature.

Designers can strategically form unfamiliar, interdisciplinary teams when the goal is to foster exploratory dialog and perspective-taking.

Practically, the results offer actionable guidance for instructional design. Designers can strategically form unfamiliar, interdisciplinary teams when the goal is to promote exploratory dialog and perspective-taking. Conversely, familiar, intra-disciplinary teams may be optimal when the goal is efficient task execution and reliable workload sharing. Understanding how personality (dis)similarity and familiarity jointly operate can guide pairing strategies to optimize satisfaction and involvement.

In sum, this study shows that the effects of social composition in IVR are context-dependent, strongly shaped by the specific task and the relational context. By highlighting these interactions and their theoretical implications, the study provides a more refined foundation for both the design of collaborative immersive environments and future research on the social dynamics of learning in virtual spaces.

#### CRedit authorship contribution statement

**Lijie Zheng:** Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Shaoteng Ke:** Validation, Software, Methodology, Formal analysis, Data curation. **Yuxin Zang:** Writing – original draft, Visualization, Formal analysis, Data curation. **Linjing Sun:** Methodology, Formal analysis. **Matthew Pike:** Writing – review & editing, Validation, Supervision. **Boon Giin Lee:** Writing – review & editing, Supervision,

Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

#### Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT for proof reading and spell checking. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Appendix

See [Tables A.1–A.13](#).

#### Data availability

Data will be made available on request.

**Table A.2**

Demographics of participants in unfamiliar pairs, categorized by academic discipline (three faculties) and personality trait (extrovert or introvert). Participants are labeled Participant A (PA) and Participant B (PB) within each pair (ID: FXX). Faculty abbreviations: Business (BS), Science & Engineering (SE), and Social Sciences (SS). A ✓ denotes an intra-faculty pairing or matched personality trait.

| ID     | Age | Faculty | Personality | ID     | Age | Faculty | Personality | Intra-Faculty | Matched personality trait |
|--------|-----|---------|-------------|--------|-----|---------|-------------|---------------|---------------------------|
| U03-PA | 21  | SE      | Introvert   | U03-PB | 20  | SE      | Extrovert   | ✓             |                           |
| U04-PA | 21  | SE      | Introvert   | U04-PB | 20  | SE      | Introvert   | ✓             | ✓                         |
| U06-PA | 19  | SE      | Introvert   | U06-PB | 21  | SE      | Extrovert   | ✓             |                           |
| U12-PA | 19  | SS      | Introvert   | U12-PB | 24  | SE      | Extrovert   |               |                           |
| U14-PA | 23  | SS      | Introvert   | U14-PB | 25  | SS      | Introvert   | ✓             | ✓                         |
| U15-PA | 24  | BS      | Extrovert   | U15-PB | 24  | SE      | Extrovert   |               | ✓                         |
| U16-PA | 24  | SS      | Extrovert   | U16-PB | 23  | SS      | Extrovert   | ✓             | ✓                         |
| U17-PA | 24  | BS      | Extrovert   | U17-PB | 24  | BS      | Extrovert   | ✓             | ✓                         |
| U18-PA | 20  | BS      | Extrovert   | U18-PB | 20  | SS      | Extrovert   |               | ✓                         |
| U19-PA | 20  | BS      | Extrovert   | U19-PB | 20  | SS      | Extrovert   |               | ✓                         |
| U20-PA | 21  | BS      | Extrovert   | U20-PB | 22  | SS      | Extrovert   |               | ✓                         |
| U25-PA | 22  | SE      | Introvert   | U25-PB | 22  | SS      | Introvert   |               | ✓                         |
| U28-PA | 25  | SS      | Extrovert   | U28-PB | 25  | BS      | Extrovert   |               | ✓                         |
| U29-PA | 25  | BS      | Introvert   | U29-PB | 26  | BS      | Extrovert   | ✓             |                           |
| U30-PA | 21  | SE      | Extrovert   | U30-PB | 21  | SE      | Extrovert   | ✓             | ✓                         |
| U31-PA | 23  | BS      | Introvert   | U31-PB | 22  | SE      | Introvert   |               | ✓                         |
| U36-PA | 25  | BS      | Introvert   | U36-PB | 21  | SE      | Extrovert   |               |                           |
| U37-PA | 20  | SE      | Introvert   | U37-PB | 20  | SE      | Introvert   | ✓             | ✓                         |
| U38-PA | 22  | BS      | Extrovert   | U38-PB | 21  | BS      | Extrovert   | ✓             | ✓                         |
| U39-PA | 21  | SS      | Introvert   | U39-PB | 21  | SE      | Extrovert   |               |                           |
| U42-PA | 22  | SE      | Extrovert   | U42-PB | 22  | SE      | Extrovert   | ✓             | ✓                         |
| U43-PA | 20  | BS      | Introvert   | U43-PB | 21  | SS      | Introvert   |               | ✓                         |
| U44-PA | 20  | SE      | Extrovert   | U44-PB | 20  | SE      | Extrovert   | ✓             | ✓                         |
| U45-PA | 21  | SS      | Extrovert   | U45-PB | 22  | BS      | Extrovert   |               | ✓                         |
| U46-PA | 25  | SS      | Extrovert   | U46-PB | 24  | SS      | Extrovert   | ✓             | ✓                         |
| U53-PA | 22  | SE      | Introvert   | U53-PB | 25  | BS      | Extrovert   |               |                           |
| U54-PA | 20  | BS      | Extrovert   | U54-PB | 24  | SE      | Introvert   |               |                           |

**Table A.3**

Observation template (Marroun & Young, 2024) for assessing pair of participants social behavior and interaction during experiments at three time intervals using a 5-point Likert scale. The “maintains mutual understanding” dimension (code A1) evaluates how well participants align their perspectives between disciplines and personality types. “Dialog management” dimension (code A2) assesses the clarity and constructiveness of their communication, while “information set” dimension (code A3) examines the efficiency of the sharing of cross-disciplinary knowledge. “Reaching consensus” dimension (code A4) reveals differences in communication strategies related to personality trait. The “division of tasks” (code A5), “time management” (code A6), and “technical coordination” (code A7) dimensions measure the mechanics of cooperative work between participants. “Reciprocal interaction” dimension (code A8) quantifies the exchange of ideas and the quality of mutual support, and “orientation of individual tasks” dimension (code A9) measures the commitment of each participant to their responsibilities during collaborative participation.

| Code | Dimension                       | Participant | Evaluation 1 <sup>a</sup> | Evaluation 2 <sup>a</sup> | Evaluation 3 <sup>a</sup> |
|------|---------------------------------|-------------|---------------------------|---------------------------|---------------------------|
| A1   | Maintain mutual understanding   | A<br>B      |                           |                           |                           |
| A2   | Dialog management               | A<br>B      |                           |                           |                           |
| A3   | Information set                 | A<br>B      |                           |                           |                           |
| A4   | Reaching consensus              | A<br>B      |                           |                           |                           |
| A5   | Division of tasks               | A<br>B      |                           |                           |                           |
| A6   | Time management                 | A<br>B      |                           |                           |                           |
| A7   | Technical coordination          | A<br>B      |                           |                           |                           |
| A8   | Reciprocal interaction          | A<br>B      |                           |                           |                           |
| A9   | Orientation of individual tasks | A<br>B      |                           |                           |                           |

<sup>a</sup> 5-Likert Scale: 0 - Very Poor; 1 - Poor; 2 - Neutral; 3 - Good; 4 - Very Good.

**Table A.4**

The 14-question semi-structured interview questions, are organized into three sections. Those addressing RQ1 explore the participant's gameplay experiences and engagement levels to determine how specific collaborative elements supported the development of cross-disciplinary social interactions. Those targeting RQ2 focused on interaction patterns and teamwork dynamics, seeking to identify differences in engagement between introverted and extroverted individuals. In addition to clarifying the effects of discipline diversity and personality composition, participant feedback and reflections provided practical design recommendations for future improvements to cooperative IVR game experiences that accommodate diverse interaction preferences.

| No.                                     | Question                                                                                                                                                                                      |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part 1: Game experience and engagement  |                                                                                                                                                                                               |
| 1                                       | How would you evaluate your overall experience with this IVR game? How did your emotions change at different phases, such as starting, solving puzzles, and finishing the game?               |
| 2                                       | How did the puzzle-solving features of the game affect your engagement and willingness to interact with your teammate?                                                                        |
| 3                                       | Which specific aspects of the game encourage or discourage cooperation? Could you give examples and elaborate on their significance?                                                          |
| 4                                       | How did the various mini-games affect your engagement with your teammate? Which mini-game offered the greatest opportunities for communication or cooperation, and why?                       |
| 5                                       | Could you elaborate on instances where you experienced different emotions such as frustration, excitement, or achievement during the game? How did these feelings influence your teamwork?    |
| Part 2: Social interaction and teamwork |                                                                                                                                                                                               |
| 6                                       | How would you describe your initial comfort in collaborating with your teammate, regardless of whether they were a stranger or friend? How did this comfort level change throughout the game? |
| 7                                       | Can you identify particular moments in the game where you were motivated to communicate or collaborate? What happened during those times?                                                     |
| 8                                       | In what ways did your willingness to interact with your teammate evolve as the game advanced? What factors influenced these variations?                                                       |
| 9                                       | Did you face any difficulties in communicating with your teammate during puzzle solving? How did you manage or resolve these issues?                                                          |
| 10                                      | How would you evaluate the effectiveness of your teamwork in this IVR game? What factors led to successful or less effective cooperative?                                                     |
| 11                                      | Did you depend on your teammate for any aspect during the game, and how did this affect your view on cooperation and teamwork?                                                                |
| Part 3: General reflections             |                                                                                                                                                                                               |
| 12                                      | What did you learn about your communication or cooperative skills from this IVR gaming experience?                                                                                            |
| 13                                      | What impact do you think this IVR gameplay has had on your future willingness to engage with strangers? Could you elaborate on your reasoning?                                                |
| 14                                      | Would you be interested in adding further insights about your experiences with this IVR game?                                                                                                 |

**Table A.5**

Descriptive statistics for the GEQ\_SPM, IMMS, SOCS, and CLSS\_CM questionnaires by familiarity, academic discipline, and personality trait.  $\mu$  indicates the mean score for each group;  $\sigma$  denotes the standard deviation;  $N$  denotes the sample size (number of pairs) per condition.

| Familiarity                                                                         | Group               | Statistic | GEQ_SPM-EM | GEQ_SPM-NF | GEQ_SPM-BI | IMMS-A | IMMS-R | IMMS-C | IMMS-S | SOCS  | CLSS_CM |
|-------------------------------------------------------------------------------------|---------------------|-----------|------------|------------|------------|--------|--------|--------|--------|-------|---------|
| Moderator: Academic discipline (Intra- vs. Inter-Faculty)                           |                     |           |            |            |            |        |        |        |        |       |         |
| Unfamiliar                                                                          | Intra-Faculty       | $\mu$     | 3.160      | 0.892      | 3.154      | 3.240  | 2.524  | 2.596  | 3.100  | 2.938 | 2.971   |
|                                                                                     |                     | $\sigma$  | 0.264      | 0.345      | 0.340      | 0.274  | 0.334  | 0.394  | 0.350  | 0.249 | 0.290   |
|                                                                                     |                     | $N$       | 13         | 13         | 13         | 13     | 13     | 13     | 13     | 13    | 13      |
| Unfamiliar                                                                          | Inter-Faculty       | $\mu$     | 3.357      | 0.964      | 3.238      | 3.286  | 2.406  | 2.397  | 3.286  | 3.093 | 3.067   |
|                                                                                     |                     | $\sigma$  | 0.379      | 0.322      | 0.409      | 0.225  | 0.482  | 0.424  | 0.342  | 0.303 | 0.244   |
|                                                                                     |                     | $N$       | 14         | 14         | 14         | 14     | 14     | 14     | 14     | 14    | 14      |
| Familiar                                                                            | Intra-Faculty       | $\mu$     | 3.133      | 0.993      | 3.061      | 3.121  | 2.688  | 3.054  | 3.067  | 3.209 | 3.133   |
|                                                                                     |                     | $\sigma$  | 0.354      | 0.381      | 0.441      | 0.339  | 0.420  | 0.371  | 0.280  | 0.311 | 0.392   |
|                                                                                     |                     | $N$       | 12         | 12         | 12         | 12     | 12     | 12     | 12     | 12    | 12      |
| Familiar                                                                            | Inter-Faculty       | $\mu$     | 3.132      | 0.842      | 2.819      | 3.047  | 2.318  | 3.161  | 3.017  | 3.039 | 3.188   |
|                                                                                     |                     | $\sigma$  | 0.246      | 0.277      | 0.322      | 0.327  | 0.276  | 0.279  | 0.377  | 0.272 | 0.321   |
|                                                                                     |                     | $N$       | 15         | 15         | 15         | 15     | 15     | 15     | 15     | 15    | 15      |
| Moderator: Personality trait (Introvert–Introvert, Extrovert–Extrovert, Mismatched) |                     |           |            |            |            |        |        |        |        |       |         |
| Unfamiliar                                                                          | Introvert–Introvert | $\mu$     | 3.194      | 0.800      | 3.111      | 3.271  | 2.448  | 2.500  | 3.117  | 2.939 | 2.938   |
|                                                                                     |                     | $\sigma$  | 0.401      | 0.270      | 0.351      | 0.219  | 0.253  | 0.388  | 0.276  | 0.233 | 0.269   |
|                                                                                     |                     | $N$       | 6          | 6          | 6          | 6      | 6      | 6      | 6      | 6     | 6       |
| Unfamiliar                                                                          | Extrovert–Extrovert | $\mu$     | 3.222      | 1.008      | 3.285      | 3.323  | 2.599  | 2.536  | 3.342  | 3.069 | 3.078   |
|                                                                                     |                     | $\sigma$  | 0.302      | 0.336      | 0.305      | 0.255  | 0.394  | 0.478  | 0.309  | 0.318 | 0.241   |
|                                                                                     |                     | $N$       | 13         | 13         | 13         | 13     | 13     | 13     | 13     | 13    | 13      |
| Unfamiliar                                                                          | Mismatched          | $\mu$     | 3.361      | 0.911      | 3.139      | 3.181  | 2.292  | 2.431  | 3.056  | 3.004 | 3.000   |
|                                                                                     |                     | $\sigma$  | 0.344      | 0.351      | 0.320      | 0.247  | 0.485  | 0.362  | 0.399  | 0.276 | 0.300   |
|                                                                                     |                     | $N$       | 8          | 8          | 8          | 8      | 8      | 8      | 8      | 8     | 8       |
| Familiar                                                                            | Introvert–Introvert | $\mu$     | 3.028      | 0.817      | 2.792      | 2.885  | 2.406  | 3.052  | 2.933  | 3.006 | 3.229   |
|                                                                                     |                     | $\sigma$  | 0.211      | 0.301      | 0.349      | 0.313  | 0.262  | 0.351  | 0.365  | 0.318 | 0.310   |
|                                                                                     |                     | $N$       | 5          | 5          | 5          | 5      | 5      | 5      | 5      | 5     | 5       |
| Familiar                                                                            | Extrovert–Extrovert | $\mu$     | 3.167      | 0.967      | 2.843      | 3.076  | 2.500  | 3.069  | 3.022  | 3.215 | 3.201   |
|                                                                                     |                     | $\sigma$  | 0.256      | 0.365      | 0.385      | 0.321  | 0.393  | 0.346  | 0.349  | 0.354 | 0.291   |
|                                                                                     |                     | $N$       | 9          | 9          | 9          | 9      | 9      | 9      | 9      | 9     | 9       |
| Familiar                                                                            | Mismatched          | $\mu$     | 3.160      | 0.950      | 3.118      | 3.198  | 2.599  | 3.151  | 3.117  | 3.136 | 3.089   |
|                                                                                     |                     | $\sigma$  | 0.376      | 0.350      | 0.404      | 0.313  | 0.466  | 0.326  | 0.276  | 0.240 | 0.426   |
|                                                                                     |                     | $N$       | 13         | 13         | 13         | 13     | 13     | 13     | 13     | 13    | 13      |

GEQ\_SPM: EM — Empathy; NF — Negative Feelings; BI — Behavioral Involvement.

IMMS: A — Attention; R — Relevance; C — Confidence; S — Satisfaction.

**Table A.6**

Levene's tests of homogeneity of variance for the questionnaire scales.

| Measurement | FM $\times$ AD: $p$ | FM $\times$ PT: $p$ |
|-------------|---------------------|---------------------|
| GEQ_SPM-EM  | 0.139               | 0.111               |
| GEQ_SPM-NF  | 0.555               | 0.826               |
| GEQ_SPM-BI  | 0.360               | 0.955               |
| IMMS-A      | 0.080               | 0.677               |
| IMMS-R      | 0.105               | 0.165               |
| IMMS-C      | 0.302               | 0.381               |
| IMMS-S      | 0.681               | 0.793               |
| SOCS        | 0.582               | 0.529               |
| CLSS_CM     | 0.159               | 0.084               |

**Table A.7**

Fisher's LSD post-hoc pairwise comparisons of GEQ-SPM-BI, IMMS-A, and IMMS-S for the interactions between familiarity (FM) and personality trait (PT).  $\Delta$  represents the mean difference between groups; SE is the standard error of the main difference;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size.

| Measurement | Condition | Comparison                          | $\Delta$ | SE    | $p$       | 95% CI |        | Cohen's $d$ |
|-------------|-----------|-------------------------------------|----------|-------|-----------|--------|--------|-------------|
|             |           |                                     |          |       |           | LL     | UL     |             |
| GEQ-SPM-BI  | FM        | Familiar: E-E vs. I-I               | 0.051    | 0.132 | 0.701     | -0.211 | 0.313  | 0.137       |
|             |           | Familiar: E-E vs. Mismatched        | 0.275    | 0.111 | 0.014*    | -0.495 | -0.056 | 0.694       |
|             |           | Familiar: I-I vs. Mismatched        | 0.326    | 0.125 | 0.011*    | 0.078  | 0.575  | 0.842       |
|             |           | Unfamiliar: E-E vs. I-I             | 0.174    | 0.125 | 0.169     | -0.075 | 0.422  | 0.542       |
|             |           | Unfamiliar: E-E vs. Mismatched      | 0.146    | 0.111 | 0.190     | -0.074 | 0.365  | 0.469       |
|             |           | Unfamiliar: I-I vs. Mismatched      | 0.028    | 0.132 | 0.834     | -0.235 | 0.290  | 0.084       |
|             | PT        | E-E: Familiar vs. Unfamiliar        | 0.442    | 0.111 | <0.001*** | -0.662 | -0.223 | 1.296       |
|             |           | I-I: Familiar vs. Unfamiliar        | 0.319    | 0.145 | 0.030*    | -0.607 | -0.032 | 0.911       |
|             |           | Mismatched: Familiar vs. Unfamiliar | 0.021    | 0.111 | 0.851     | -0.240 | 0.199  | 0.057       |
| IMMS-A      | FM        | Familiar: E-E vs. I-I               | 0.191    | 0.105 | 0.073     | -0.018 | 0.400  | 0.601       |
|             |           | Familiar: E-E vs. Mismatched        | 0.122    | 0.088 | 0.171     | -0.296 | 0.053  | 0.386       |
|             |           | Familiar: I-I vs. Mismatched        | 0.313    | 0.100 | 0.002**   | -0.511 | -0.114 | 1.000       |
|             |           | Unfamiliar: E-E vs. I-I             | 0.052    | 0.100 | 0.603     | -0.146 | 0.250  | 0.214       |
|             |           | Unfamiliar: E-E vs. Mismatched      | 0.142    | 0.088 | 0.109     | -0.032 | 0.317  | 0.566       |
|             |           | Unfamiliar: I-I vs. Mismatched      | 0.090    | 0.105 | 0.393     | -0.119 | 0.299  | 0.380       |
|             | PT        | E-E: Familiar vs. Unfamiliar        | 0.247    | 0.088 | 0.006**   | -0.421 | -0.072 | 0.867       |
|             |           | I-I: Familiar vs. Unfamiliar        | 0.385    | 0.115 | 0.001**   | -0.614 | -0.157 | 1.430       |
|             |           | Mismatched: Familiar vs. Unfamiliar | 0.017    | 0.088 | 0.844     | -0.157 | 0.192  | 0.059       |
| IMMS-S      | FM        | Familiar: E-E vs. I-I               | 0.089    | 0.123 | 0.470     | -0.154 | 0.332  | 0.251       |
|             |           | Familiar: E-E vs. Mismatched        | 0.094    | 0.103 | 0.359     | -0.298 | 0.109  | 0.306       |
|             |           | Familiar: I-I vs. Mismatched        | 0.183    | 0.116 | 0.118     | -0.414 | 0.047  | 0.599       |
|             |           | Unfamiliar: E-E vs. I-I             | 0.225    | 0.116 | 0.056     | -0.006 | 0.456  | 0.755       |
|             |           | Unfamiliar: E-E vs. Mismatched      | 0.286    | 0.103 | 0.006**   | 0.083  | 0.489  | 0.819       |
|             |           | Unfamiliar: I-I vs. Mismatched      | 0.061    | 0.123 | 0.619     | -0.182 | 0.304  | 0.172       |
|             | PT        | E-E: Familiar vs. Unfamiliar        | 0.319    | 0.103 | 0.002**   | -0.523 | -0.116 | 0.979       |
|             |           | I-I: Familiar vs. Unfamiliar        | 0.183    | 0.134 | 0.175     | -0.083 | 0.450  | 0.568       |
|             |           | Mismatched: Familiar vs. Unfamiliar | 0.061    | 0.103 | 0.553     | -0.142 | 0.264  | 0.183       |

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

E-E — Extrovert-Extrovert; I-I — Introvert-Introvert.

**Table A.8**

EMM post-hoc pairwise comparisons of observational codes by personality trait.  $\Delta$  represents the mean difference between groups; SE is the standard error of the main difference;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size.

| Code | Comparison         | $\Delta$ | SE    | $p$       | 95% CI |        | Cohen's $d$ |
|------|--------------------|----------|-------|-----------|--------|--------|-------------|
|      |                    |          |       |           | LL     | UL     |             |
| A1   | E-E vs. I-I        | 0.040    | 0.164 | 0.784     | -0.370 | 0.280  | 0.061       |
|      | E-E vs. Mismatched | 0.370    | 0.143 | 0.009**   | 0.090  | 0.650  | 0.568       |
|      | I-I vs. Mismatched | 0.420    | 0.167 | 0.012*    | 0.090  | 0.740  | 0.644       |
| A2   | E-E vs. I-I        | 0.550    | 0.156 | <0.001*** | -0.860 | -0.250 | 0.889       |
|      | E-E vs. Mismatched | 0.030    | 0.136 | 0.820     | -0.240 | 0.300  | 0.048       |
|      | I-I vs. Mismatched | 0.580    | 0.159 | <0.001*** | 0.270  | 0.900  | 0.937       |
| A4   | E-E vs. I-I        | 0.170    | 0.167 | 0.308     | -0.500 | 0.160  | 0.177       |
|      | E-E vs. Mismatched | 0.240    | 0.145 | 0.104     | -0.050 | 0.520  | 0.249       |
|      | I-I vs. Mismatched | 0.410    | 0.170 | 0.017*    | -0.740 | -0.070 | 0.426       |
| A5   | E-E vs. I-I        | 0.320    | 0.141 | 0.025*    | -0.590 | -0.040 | 0.571       |
|      | E-E vs. Mismatched | 0.300    | 0.123 | 0.013*    | -0.540 | -0.060 | 0.536       |
|      | I-I vs. Mismatched | 0.010    | 0.143 | 0.942     | -0.270 | 0.290  | 0.018       |
| A6   | E-E vs. I-I        | 0.260    | 0.121 | 0.030*    | -0.500 | -0.030 | 0.538       |
|      | E-E vs. Mismatched | 0.130    | 0.106 | 0.213     | -0.080 | 0.340  | 0.270       |
|      | I-I vs. Mismatched | 0.400    | 0.124 | 0.001**   | 0.150  | 0.640  | 0.807       |
| A7   | E-E vs. I-I        | 0.450    | 0.162 | 0.005**   | -0.770 | -0.140 | 0.702       |
|      | E-E vs. Mismatched | 0.120    | 0.141 | 0.393     | -0.400 | 0.160  | 0.187       |
|      | I-I vs. Mismatched | 0.330    | 0.165 | 0.043*    | -0.660 | -0.010 | 0.515       |
| A8   | E-E vs. I-I        | 0.010    | 0.155 | 0.945     | -0.290 | 0.310  | 0.016       |
|      | E-E vs. Mismatched | 0.320    | 0.135 | 0.017*    | 0.060  | 0.590  | 0.518       |
|      | I-I vs. Mismatched | 0.310    | 0.158 | 0.048*    | 0.000  | 0.620  | 0.503       |
| A9   | E-E vs. I-I        | 0.430    | 0.145 | 0.003**   | -0.720 | -0.150 | 0.749       |
|      | E-E vs. Mismatched | 0.010    | 0.126 | 0.906     | -0.260 | 0.230  | 0.017       |
|      | I-I vs. Mismatched | 0.420    | 0.147 | 0.005**   | 0.130  | 0.710  | 0.732       |

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

E-E — extrovert-extrovert; I-I — Introvert-Introvert.

**Table A.9**

EMM post-hoc pairwise comparisons of observational codes A6 and A9 for the interactions between familiarity (FM) and personality trait (PT).  $\Delta$  represents the mean difference between groups; SE is the standard error of the main difference;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size.

| Code | Condition | Comparison                          | $\Delta$ | SE    | $p$       | 95% CI |        | Cohen's $d$ |
|------|-----------|-------------------------------------|----------|-------|-----------|--------|--------|-------------|
|      |           |                                     |          |       |           | LL     | UL     |             |
| A6   | FM        | Familiar: E-E vs. I-I               | 0.190    | 0.177 | 0.273     | -0.540 | 0.150  | 0.389       |
|      |           | Familiar: E-E vs. Mismatched        | 0.110    | 0.148 | 0.454     | -0.400 | 0.180  | 0.226       |
|      |           | Familiar: I-I vs. Mismatched        | 0.080    | 0.168 | 0.620     | -0.410 | 0.250  | 0.164       |
|      |           | Unfamiliar: E-E vs. I-I             | 0.330    | 0.166 | 0.045*    | -0.660 | -0.010 | 0.681       |
|      |           | Unfamiliar: E-E vs. Mismatched      | 0.370    | 0.151 | 0.013*    | 0.080  | 0.670  | 0.764       |
|      |           | Unfamiliar: I-I vs. Mismatched      | 0.710    | 0.182 | <0.001*** | 0.350  | 1.060  | 1.467       |
|      | PT        | E-E: Familiar vs. Unfamiliar        | 0.440    | 0.146 | 0.002**   | -0.730 | -0.160 | 0.911       |
|      |           | I-I: Familiar vs. Unfamiliar        | 0.580    | 0.194 | 0.003**   | 0.200  | 0.960  | 1.201       |
|      |           | Mismatched: Familiar vs. Unfamiliar | 0.040    | 0.154 | 0.786     | -0.260 | 0.340  | 0.083       |
| A9   | FM        | Familiar: E-E vs. I-I               | 0.470    | 0.211 | 0.025*    | -0.890 | -0.060 | 0.811       |
|      |           | Familiar: E-E vs. Mismatched        | 0.350    | 0.177 | 0.049*    | -0.690 | 0.000  | 0.604       |
|      |           | Familiar: I-I vs. Mismatched        | 0.120    | 0.200 | 0.533     | -0.270 | 0.520  | 0.207       |
|      |           | Unfamiliar: E-E vs. I-I             | 0.390    | 0.198 | 0.048*    | -0.780 | 0.000  | 0.673       |
|      |           | Unfamiliar: E-E vs. Mismatched      | 0.320    | 0.180 | 0.078     | -0.040 | 0.670  | 0.553       |
|      |           | Unfamiliar: I-I vs. Mismatched      | 0.710    | 0.216 | 0.001**   | 0.280  | 1.130  | 1.226       |
|      | PT        | E-E: Familiar vs. Unfamiliar        | 0.410    | 0.174 | 0.017*    | -0.760 | -0.070 | 0.711       |
|      |           | Mismatched: Familiar vs. Unfamiliar | 0.250    | 0.183 | 0.172     | -0.110 | 0.610  | 0.433       |
|      |           | I-I: Familiar vs. Unfamiliar        | 0.330    | 0.231 | 0.150     | -0.790 | 0.120  | 0.571       |

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

E-E — Extrovert-Extrovert; I-I — Introvert-Introvert.

**Table A.10**

Generalized linear model (GLM) analysis of familiarity (FM) and academic discipline (AD), including their interaction (FM  $\times$  AD), as well as familiarity (FM) and personality trait (PT), including their interaction (FM  $\times$  PT). Wald Chi-Square is the test statistic; df indicates degrees of freedom;  $p$  denotes statistical significance; and  $\eta^2 p$  represents the effect size, reflecting the proportion of total variance explained by each effect.

| Code | Source         | Wald Chi-Square | df | $p$       | $\eta^2 p$ | Source         | Wald Chi-Square | df | $p$       | $\eta^2 p$ |
|------|----------------|-----------------|----|-----------|------------|----------------|-----------------|----|-----------|------------|
| A1   | FM             | 6.657           | 1  | 0.010*    | 0.058      | FM             | 6.584           | 1  | 0.010*    | 0.057      |
|      | AD             | 3.041           | 1  | 0.081     | 0.027      | PT             | 8.993           | 2  | 0.011*    | 0.077      |
|      | FM $\times$ AD | 0.328           | 1  | 0.567     | 0.003      | F $\times$ PT  | 3.120           | 2  | 0.210     | 0.028      |
| A2   | FM             | 1.311           | 1  | 0.252     | 0.012      | FM             | 0.656           | 1  | 0.418     | 0.006      |
|      | AD             | 4.806           | 1  | 0.028*    | 0.043      | PT             | 15.974          | 2  | <0.001*** | 0.129      |
|      | FM $\times$ AD | 0.118           | 1  | 0.731     | 0.001      | FM $\times$ PT | 1.600           | 2  | 0.449     | 0.015      |
| A3   | FM             | 1.808           | 1  | 0.179     | 0.016      | FM             | 1.236           | 1  | 0.266     | 0.011      |
|      | AD             | 5.233           | 1  | 0.022*    | 0.046      | PT             | 1.800           | 2  | 0.407     | 0.016      |
|      | FM $\times$ AD | 1.859           | 1  | 0.173     | 0.017      | FM $\times$ PT | 0.074           | 2  | 0.964     | 0.001      |
| A4   | FM             | 5.076           | 1  | 0.024*    | 0.045      | FM             | 5.575           | 1  | 0.018*    | 0.049      |
|      | AD             | 0.151           | 1  | 0.697     | 0.001      | PT             | 6.100           | 2  | 0.047*    | 0.053      |
|      | FM $\times$ AD | 0.151           | 1  | 0.697     | 0.001      | FM $\times$ PT | 2.310           | 2  | 0.315     | 0.021      |
| A5   | FM             | 5.880           | 1  | 0.015*    | 0.052      | FM             | 4.167           | 1  | 0.041*    | 0.037      |
|      | AD             | 0.192           | 1  | 0.661     | 0.002      | PT             | 7.959           | 2  | 0.019*    | 0.069      |
|      | FM $\times$ AD | 0.780           | 1  | 0.377     | 0.007      | FM $\times$ PT | 5.113           | 2  | 0.078     | 0.045      |
| A6   | FM             | 11.435          | 1  | <0.001*** | 0.096      | FM             | 11.776          | 1  | <0.001*** | 0.098      |
|      | AD             | 8.626           | 1  | 0.003**   | 0.074      | PT             | 10.270          | 2  | 0.006**   | 0.087      |
|      | FM $\times$ AD | 3.173           | 1  | 0.075     | 0.029      | FM $\times$ PT | 8.060           | 2  | 0.018*    | 0.069      |
| A7   | FM             | 0.270           | 1  | 0.603     | 0.002      | FM             | 0.097           | 1  | 0.755     | 0.001      |
|      | AD             | 0.716           | 1  | 0.398     | 0.007      | PT             | 7.791           | 2  | 0.019*    | 0.069      |
|      | FM $\times$ AD | 0.155           | 1  | 0.694     | 0.001      | FM $\times$ PT | 2.742           | 2  | 0.254     | 0.025      |
| A8   | FM             | 10.552          | 1  | 0.001**   | 0.089      | FM             | 8.861           | 1  | 0.003**   | 0.076      |
|      | AD             | 0.001           | 1  | 0.977     | 0.000      | PT             | 6.751           | 2  | 0.034*    | 0.059      |
|      | FM $\times$ AD | 1.440           | 1  | 0.230     | 0.013      | FM $\times$ PT | 3.339           | 2  | 0.188     | 0.030      |
| A9   | FM             | 1.582           | 1  | 0.209     | 0.014      | FM             | 2.115           | 1  | 0.146     | 0.019      |
|      | AD             | 0.009           | 1  | 0.925     | 0.000      | PT             | 10.400          | 2  | 0.006**   | 0.088      |
|      | FM $\times$ AD | 0.001           | 1  | 0.982     | 0.000      | FM $\times$ PT | 7.726           | 2  | 0.021*    | 0.067      |

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

**Table A.11**

Two-way ANOVA analysis of familiarity (FM), academic discipline (AD), and their interactions (FM  $\times$  AD). Type III Sum of Squares quantifies the variance attributed to each effect; df denotes degrees of freedom; Mean Square is the sum of squares divided by its df;  $F$  represents the ratio of mean squares ( $F$ -statistic);  $p$  indicates statistical significance; and  $\eta^2 p$  indicates the effect size, reflecting the proportion of total variance explained by each effect.

| Measurement | Source         | Type III sum of squares | df | Mean square | $F$    | $p$       | $\eta^2 p$ |
|-------------|----------------|-------------------------|----|-------------|--------|-----------|------------|
| GEQ_SPM-EM  | FM             | 0.426                   | 1  | 0.426       | 4.164  | 0.044*    | 0.038      |
|             | AD             | 0.256                   | 1  | 0.256       | 2.504  | 0.117     | 0.023      |
|             | FM $\times$ AD | 0.264                   | 1  | 0.264       | 2.575  | 0.112     | 0.024      |
| GEQ_SPM-NF  | FM             | 0.003                   | 1  | 0.003       | 0.028  | 0.868     | 0.000      |
|             | AD             | 0.043                   | 1  | 0.043       | 0.377  | 0.540     | 0.004      |
|             | FM $\times$ AD | 0.335                   | 1  | 0.335       | 2.970  | 0.088     | 0.028      |
| GEQ_SPM-BI  | FM             | 1.753                   | 1  | 1.753       | 13.519 | <0.001*** | 0.115      |
|             | AD             | 0.166                   | 1  | 0.166       | 1.281  | 0.260     | 0.012      |
|             | FM $\times$ AD | 0.712                   | 1  | 0.712       | 5.491  | 0.021*    | 0.050      |
| IMMS-A      | FM             | 0.861                   | 1  | 0.861       | 9.913  | 0.002**   | 0.087      |
|             | AD             | 0.005                   | 1  | 0.005       | 0.063  | 0.802     | 0.001      |
|             | FM $\times$ AD | 0.095                   | 1  | 0.095       | 1.098  | 0.297     | 0.010      |
| IMMS-R      | FM             | 0.038                   | 1  | 0.038       | 0.246  | 0.621     | 0.002      |
|             | AD             | 1.594                   | 1  | 1.594       | 10.412 | 0.002**   | 0.091      |
|             | FM $\times$ AD | 0.426                   | 1  | 0.426       | 2.781  | 0.098     | 0.026      |
| IMMS-C      | FM             | 10.013                  | 1  | 10.013      | 71.734 | <0.001*** | 0.408      |
|             | AD             | 0.056                   | 1  | 0.056       | 0.402  | 0.527     | 0.004      |
|             | FM $\times$ AD | 0.628                   | 1  | 0.628       | 4.501  | 0.036*    | 0.041      |
| IMMS-S      | FM             | 0.613                   | 1  | 0.613       | 5.423  | 0.022*    | 0.049      |
|             | AD             | 0.123                   | 1  | 0.123       | 1.092  | 0.298     | 0.010      |
|             | FM $\times$ AD | 0.372                   | 1  | 0.372       | 3.295  | 0.072     | 0.030      |
| SOCS        | FM             | 0.314                   | 1  | 0.314       | 3.824  | 0.053     | 0.035      |
|             | AD             | 0.002                   | 1  | 0.002       | 0.020  | 0.888     | 0.000      |
|             | FM $\times$ AD | 0.705                   | 1  | 0.705       | 8.589  | 0.004**   | 0.076      |
| CLSS_CM     | FM             | 0.536                   | 1  | 0.536       | 5.280  | 0.024*    | 0.048      |
|             | AD             | 0.151                   | 1  | 0.151       | 1.486  | 0.226     | 0.014      |
|             | FM $\times$ AD | 0.012                   | 1  | 0.012       | 0.115  | 0.736     | 0.001      |

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

**Table A.12**

Fisher's LSD post-hoc pairwise comparisons of GEQ\_SPM-BI, IMMS-C, and SOCS for the interactions between familiarity (FM) and academic discipline (AD).  $\Delta$  represents the mean difference between groups; SE is the standard error of the main difference;  $p$  indicates statistical significance; line bars denote the 95% CI bounds (lower and upper limits); and the colored bubble, based on Cohen's  $d$ , illustrates the effect size.

| Measurement | Condition | Comparison                                  | $\Delta$ | SE    | $p$       | 95% CI |        | Cohen's $d$ |
|-------------|-----------|---------------------------------------------|----------|-------|-----------|--------|--------|-------------|
|             |           |                                             |          |       |           | LL     | UL     |             |
| GEQ_SPM-BI  | AD        | Inter-Faculty: Familiar vs. Unfamiliar      | -0.419   | 0.100 | <0.001*** | -0.617 | -0.220 | 1.330       |
|             |           | Intra-Faculty: Familiar vs. Unfamiliar      | -0.093   | 0.096 | 0.339     | -0.284 | 0.099  | 0.230       |
|             | FM        | Familiar: Inter-Faculty vs. Intra-Faculty   | -0.242   | 0.099 | 0.016*    | -0.437 | -0.046 | 0.617       |
|             |           | Unfamiliar: Inter-Faculty vs. Intra-Faculty | 0.084    | 0.098 | 0.392     | -0.110 | 0.279  | 0.259       |
| IMMS-C      | AD        | Inter-Faculty: Familiar vs. Unfamiliar      | 0.764    | 0.104 | <0.001*** | 0.558  | 0.970  | 2.099       |
|             |           | Intra-Faculty: Familiar vs. Unfamiliar      | 0.458    | 0.100 | <0.001*** | 0.259  | 0.657  | 1.212       |
|             | FM        | Familiar: Inter-Faculty vs. Intra-Faculty   | 0.107    | 0.102 | 0.297     | -0.096 | 0.310  | 0.319       |
|             |           | Unfamiliar: Inter-Faculty vs. Intra-Faculty | -0.199   | 0.102 | 0.053     | -0.401 | 0.003  | 0.484       |
| SOCS        | AD        | Inter-Faculty: Familiar vs. Unfamiliar      | -0.054   | 0.080 | 0.500     | -0.212 | 0.104  | 0.186       |
|             |           | Intra-Faculty: Familiar vs. Unfamiliar      | 0.270    | 0.077 | <0.001*** | 0.118  | 0.423  | 0.951       |
|             | FM        | Familiar: Inter-Faculty vs. Intra-Faculty   | -0.170   | 0.078 | 0.033*    | -0.326 | -0.014 | 0.576       |
|             |           | Unfamiliar: Inter-Faculty vs. Intra-Faculty | 0.154    | 0.078 | 0.051     | 0.000  | 0.309  | 0.556       |

\* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ .

Table A.13

Two-way ANOVA analysis of familiarity (FM), personality trait (PT), and their interactions (FM × PT). Type III Sum of Squares quantifies the variance attributed to each effect; df denotes degrees of freedom; Mean Square is the sum of squares divided by its df; *F* represents the ratio of mean squares (*F*-statistic); *p* indicates statistical significance; and  $\eta^2p$  indicates the effect size, reflecting the proportion of total variance explained by each effect.

| Measurement | Source  | Type III sum of squares | df | Mean square | <i>F</i> | <i>p</i>  | $\eta^2p$ |
|-------------|---------|-------------------------|----|-------------|----------|-----------|-----------|
| GEQ_SPM-EM  | FM      | 0.497                   | 1  | 0.497       | 4.726    | 0.032*    | 0.044     |
|             | PT      | 0.341                   | 2  | 0.170       | 1.621    | 0.203     | 0.031     |
|             | FM × PT | 0.116                   | 2  | 0.058       | 0.552    | 0.577     | 0.011     |
| GEQ_SPM-NF  | FM      | 0.001                   | 1  | 0.001       | 0.005    | 0.945     | 0.000     |
|             | PT      | 0.489                   | 2  | 0.244       | 2.153    | 0.121     | 0.041     |
|             | FM × PT | 0.035                   | 2  | 0.017       | 0.154    | 0.858     | 0.003     |
| GEQ_SPM-BI  | FM      | 1.695                   | 1  | 1.695       | 13.466   | <0.001*** | 0.117     |
|             | PT      | 0.475                   | 2  | 0.238       | 1.888    | 0.157     | 0.036     |
|             | FM × PT | 0.949                   | 2  | 0.474       | 3.768    | 0.026*    | 0.069     |
| IMMS-A      | FM      | 1.046                   | 1  | 1.046       | 13.106   | <0.001*** | 0.114     |
|             | PT      | 0.254                   | 2  | 0.127       | 1.589    | 0.209     | 0.030     |
|             | FM × PT | 0.699                   | 2  | 0.349       | 4.379    | 0.015*    | 0.079     |
| IMMS-R      | FM      | 0.077                   | 1  | 0.077       | 0.471    | 0.494     | 0.005     |
|             | PT      | 0.315                   | 2  | 0.158       | 0.966    | 0.384     | 0.019     |
|             | FM × PT | 0.948                   | 2  | 0.474       | 2.904    | 0.059     | 0.054     |
| IMMS-C      | FM      | 9.028                   | 1  | 9.028       | 61.471   | <0.001*** | 0.376     |
|             | PT      | 0.011                   | 2  | 0.006       | 0.038    | 0.963     | 0.001     |
|             | FM × PT | 0.207                   | 2  | 0.103       | 0.704    | 0.497     | 0.014     |
| IMMS-S      | FM      | 0.540                   | 1  | 0.540       | 4.995    | 0.028*    | 0.047     |
|             | PT      | 0.410                   | 2  | 0.205       | 1.895    | 0.156     | 0.036     |
|             | FM × PT | 0.758                   | 2  | 0.379       | 3.507    | 0.034*    | 0.064     |
| SOCS        | FM      | 0.329                   | 1  | 0.329       | 3.812    | 0.054     | 0.036     |
|             | PT      | 0.440                   | 2  | 0.220       | 2.552    | 0.083     | 0.048     |
|             | FM × PT | 0.025                   | 2  | 0.013       | 0.146    | 0.865     | 0.003     |
| CLSS_CM     | FM      | 0.702                   | 1  | 0.702       | 6.927    | 0.010*    | 0.064     |
|             | PT      | 0.189                   | 2  | 0.094       | 0.932    | 0.397     | 0.018     |
|             | FM × PT | 0.166                   | 2  | 0.083       | 0.822    | 0.443     | 0.016     |

\**p* < .05; \*\**p* < .01; \*\*\**p* < .001.

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